Chapter 6
Timing

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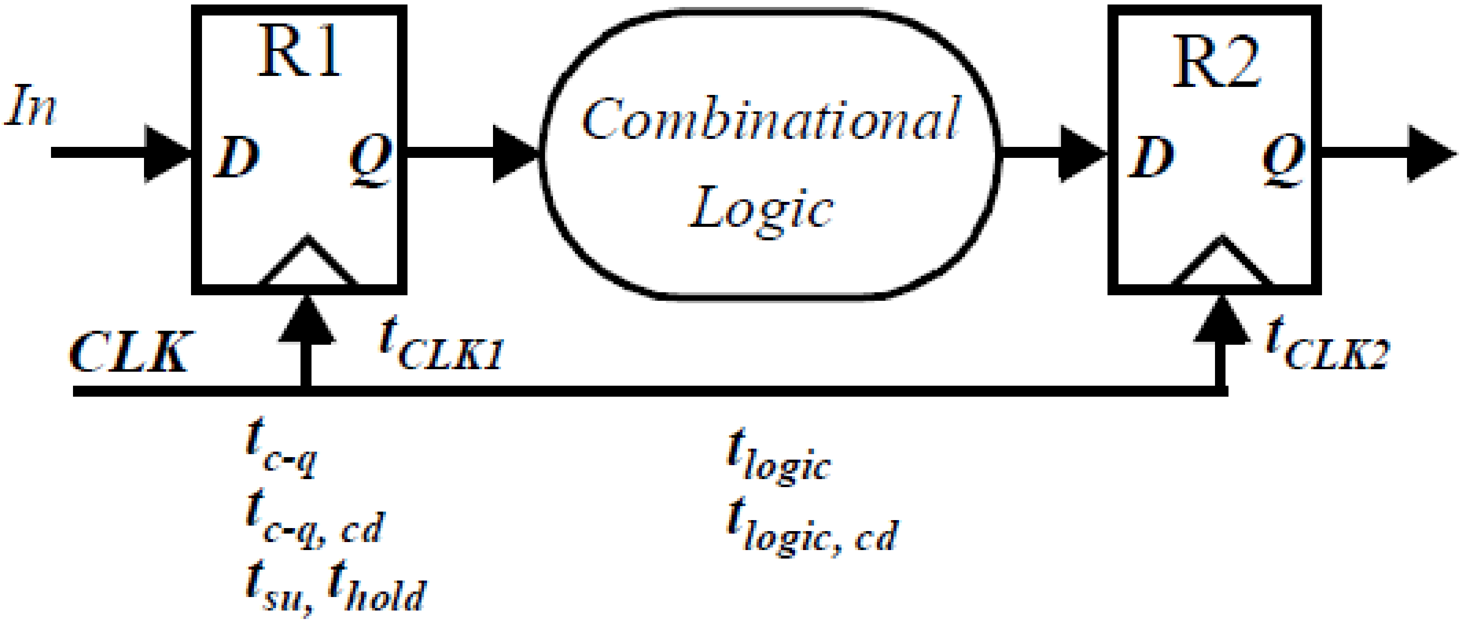
Timing Classifications

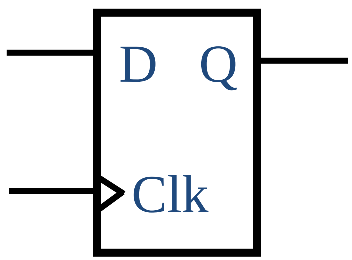
- Synchronous systems
  - All memory elements in the system are simultaneously updated using a globally distributed periodic synchronization signal (i.e., a global clock signal)
  - Functionality is ensure by strict constraints on the clock signal generation and distribution to minimize
    - Clock skew (spatial variations in clock edges)
    - Clock jitter (temporal variations in clock edges)
- Asynchronous systems
  - Self-timed (controlled) systems
  - No need for a globally distributed clock, but have asynchronous circuit overheads (handshaking logic, etc.)
- Hybrid systems
  - Synchronization between different clock domains
  - Interfacing between asynchronous and synchronous domains

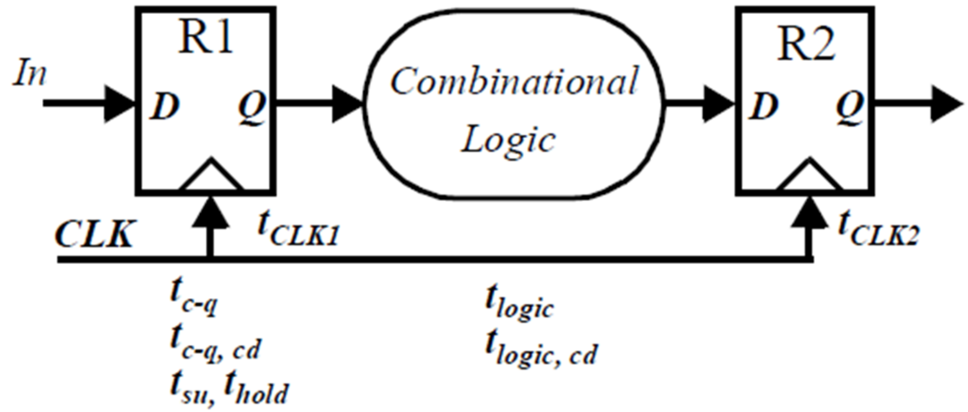
Pipelined Datapath Circuit and timing parameters

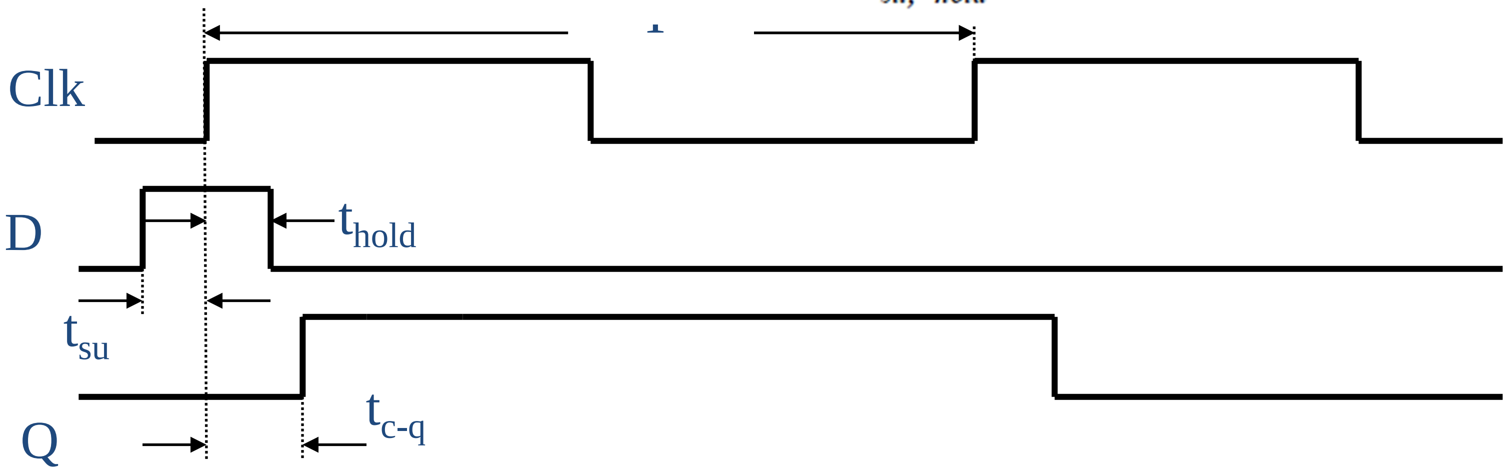


- The contamination (minimum) delay $t_{c\text{-}q,cd}$, and maximum propagation delay of the register $t_{c\text{-}q}$.
- The set-up (t_{su}) and hold time (t_{hold}) for the registers.
- The contamination delay $t_{logic,cd}$ and maximum delay t_{logic} of the combinational logic.
- *tclk1* and *tclk2*, corresponding to the position of the rising edge of the clock relative to a global reference.

Timing Definitions





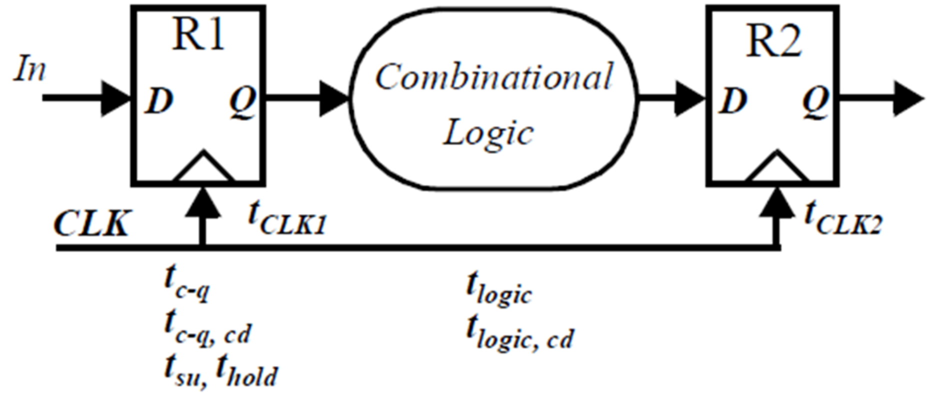


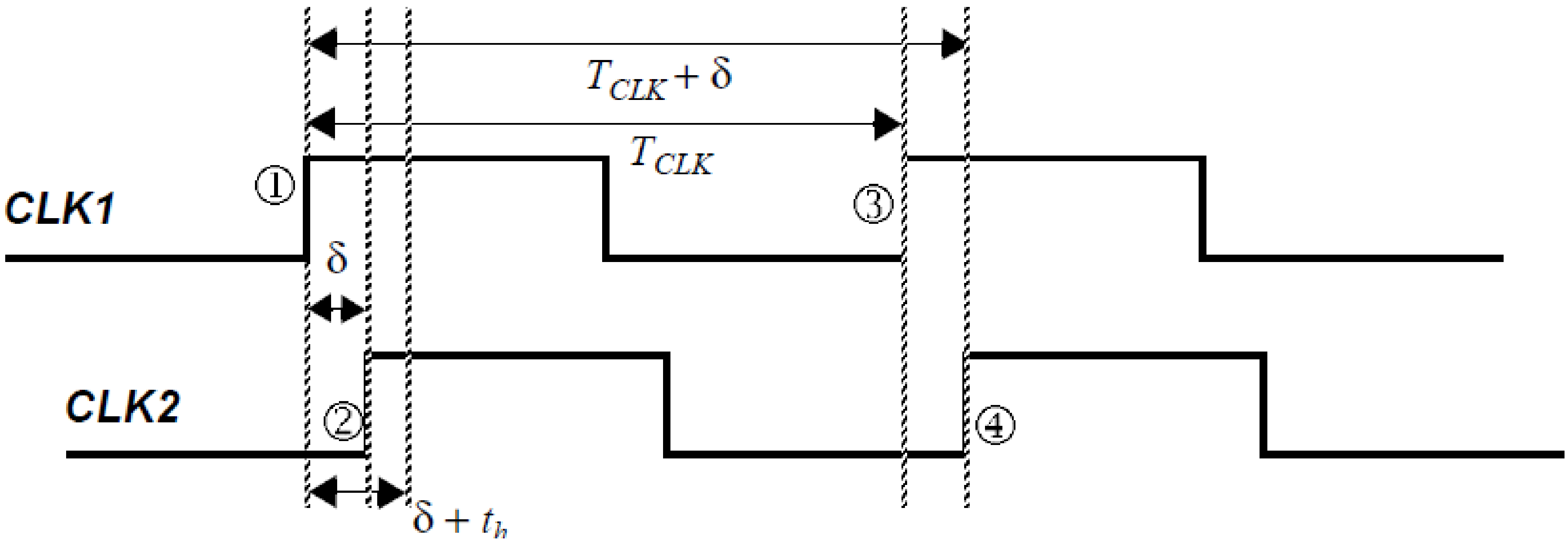
- In ideal condition,
  - The constrain of minimum clock period: $T > t_{c-q} + t_{logic} + t_{su}$
  - The hold time constrain: $t_{hold} < t_{c-q, cd} + t_{logic, cd}$
- In reality, delays can be different for rising and falling data transitions

Clock Nonidealities

- **Clock skew**
  - Spatial variation in temporally equivalent clock edges; deterministic + random, t_{SK}
- **Clock jitter**
  - Temporal variations in consecutive edges of the clock signal; modulation + random noise
  - Cycle-to-cycle (short-term) t_{JS}
  - Long term t_{JL}
- **Variation of the pulse width**
  - Related to clock jitter;
  - Important for level sensitive clocking

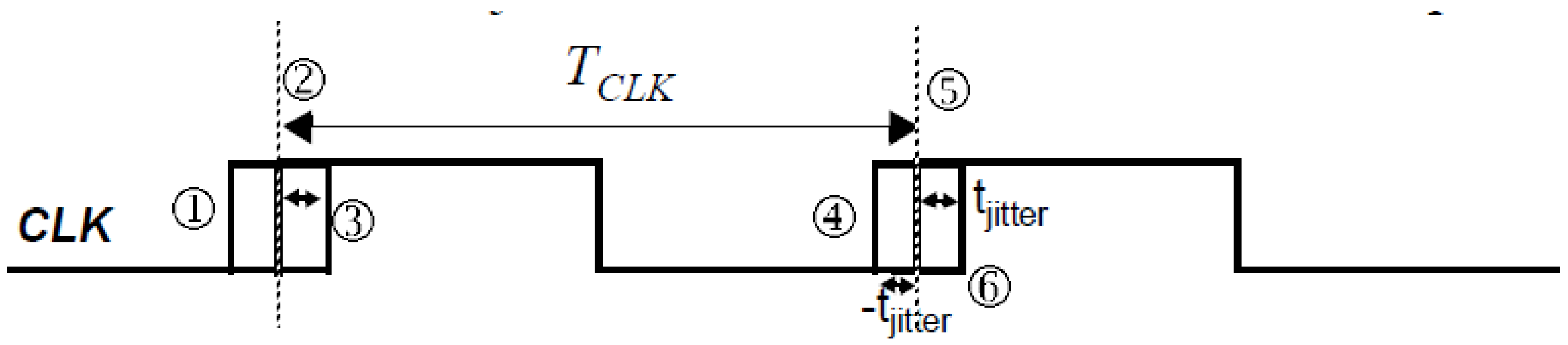
Clock Skew





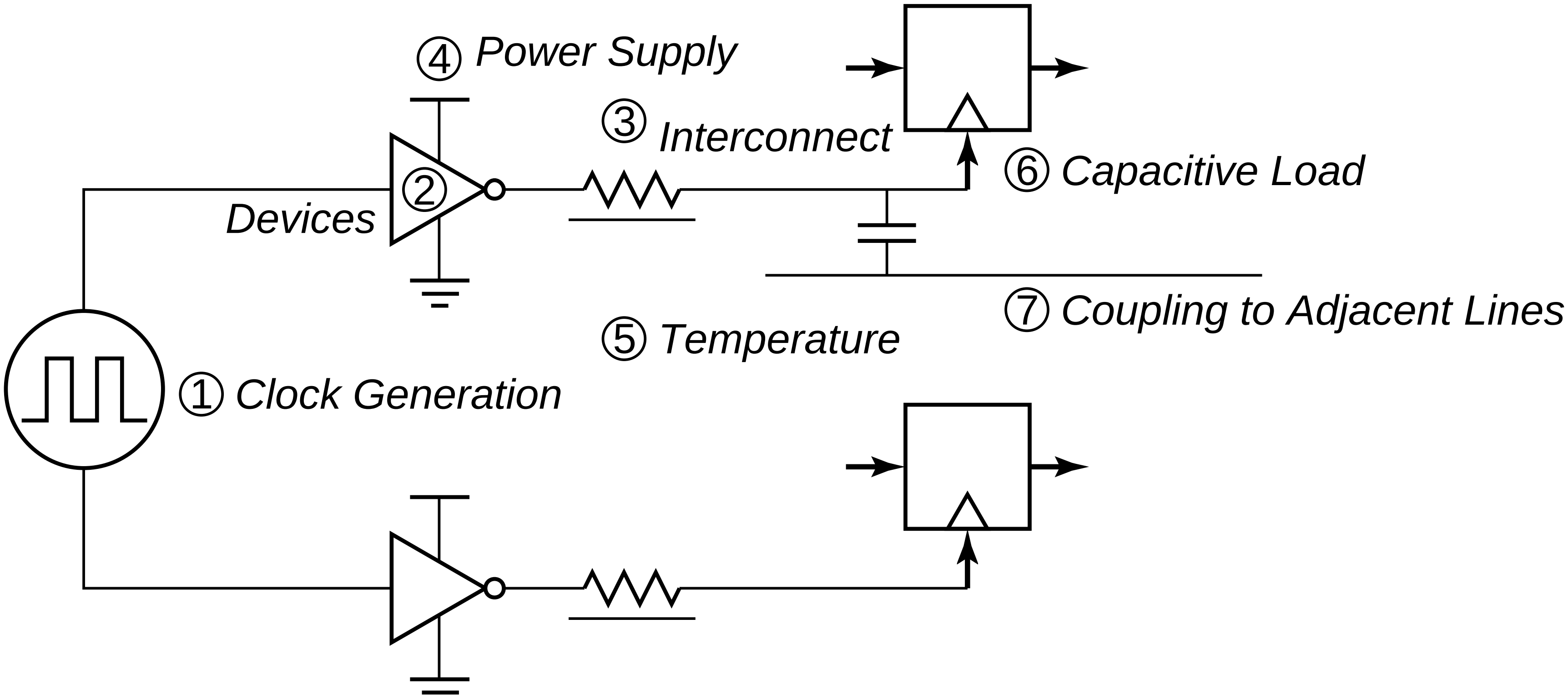
- Timing diagram to study the impact of clock skew on performance and functionality. In this sample timing diagram, $\delta > 0$.

Clock Jitter



- jitter affect the effective cycle time;
- refers to the temporal variation of the clock period at a given point;
- the clock period can reduce or expand;
- It is strictly a temporal uncertainty measure.

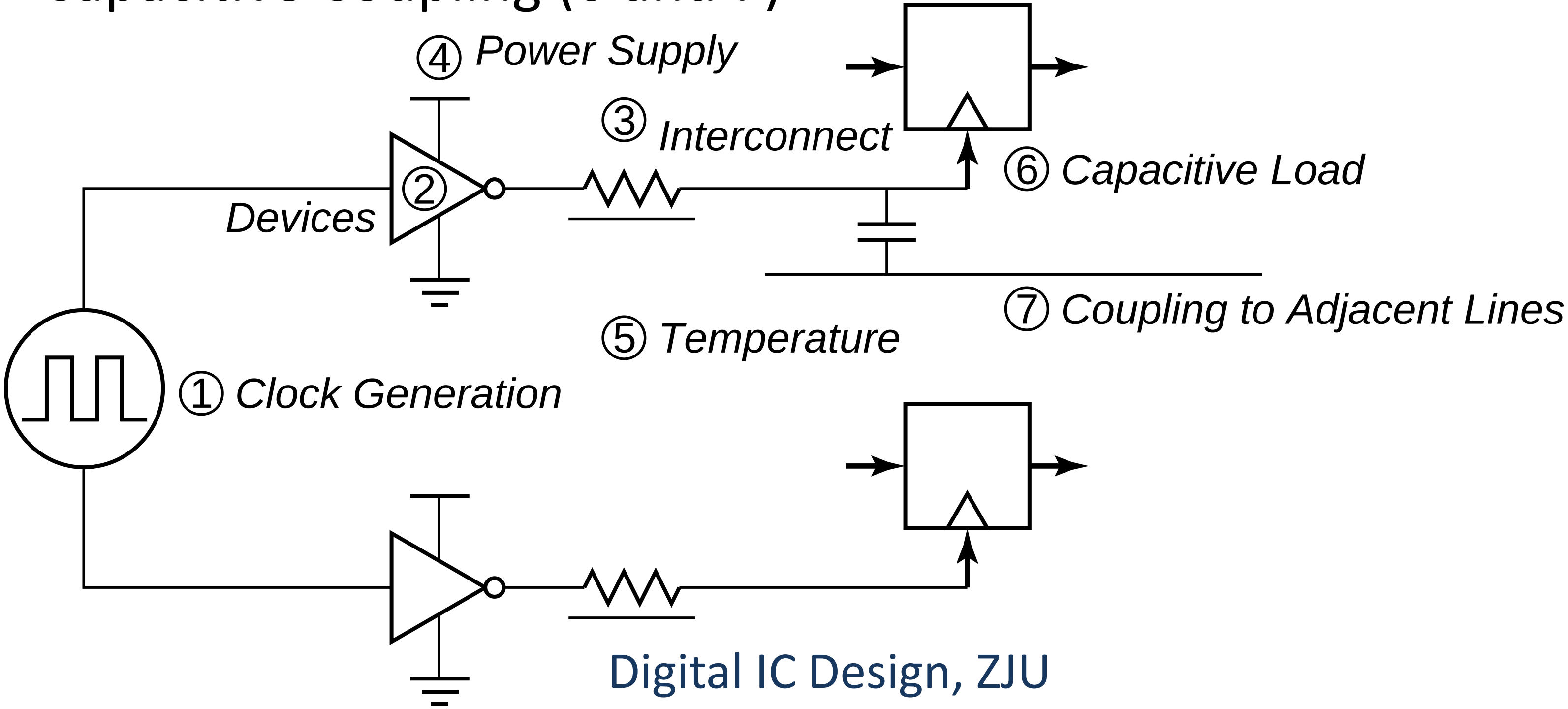
Source of the Clock Uncertainties



Sources of clock uncertainty

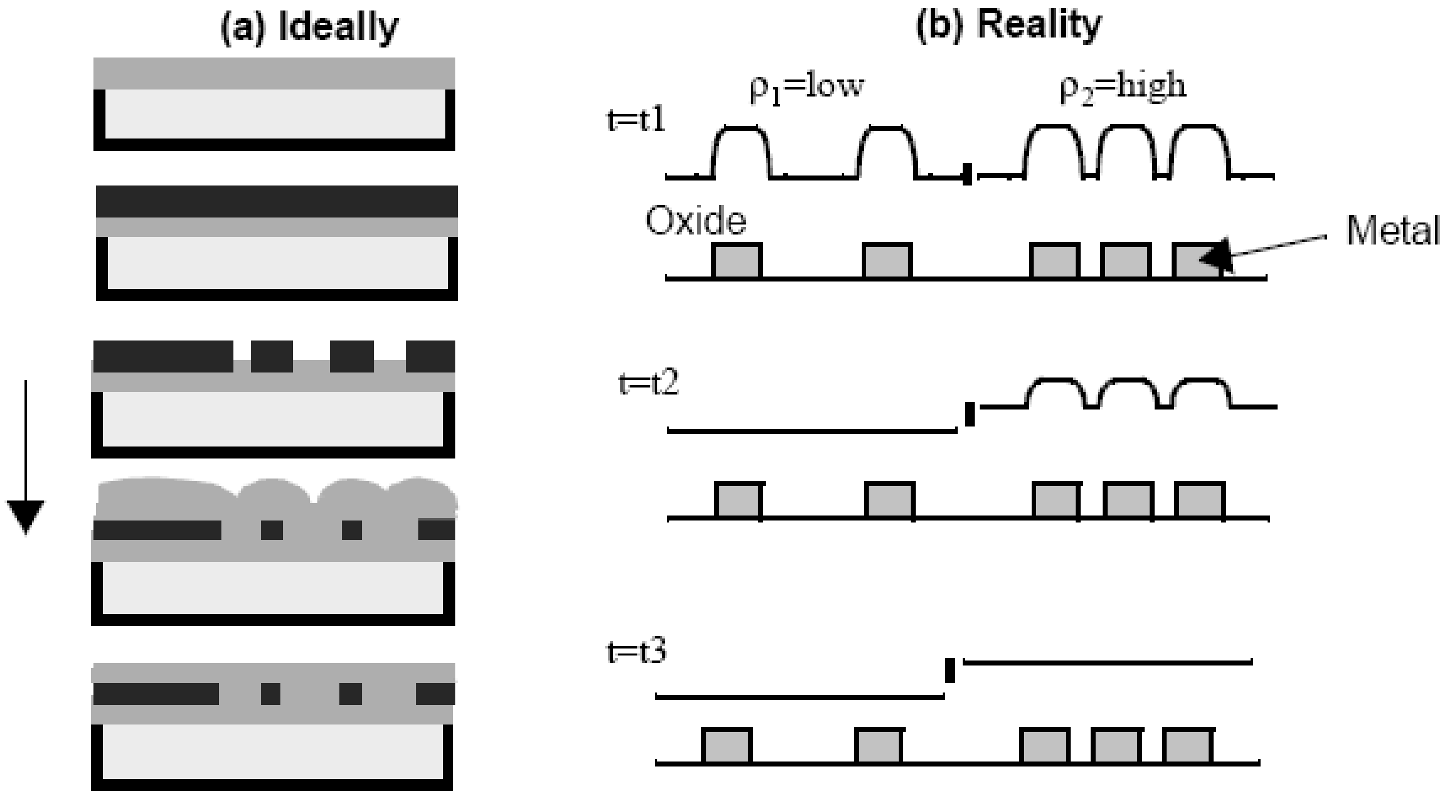
Source of the Clock Uncertainties

- Clock-Signal Generation (1)
- Manufacturing Device Variations (2)
- Interconnect Variations (3)
- Environmental Variations (4 and 5)
- Capacitive Coupling (6 and 7)

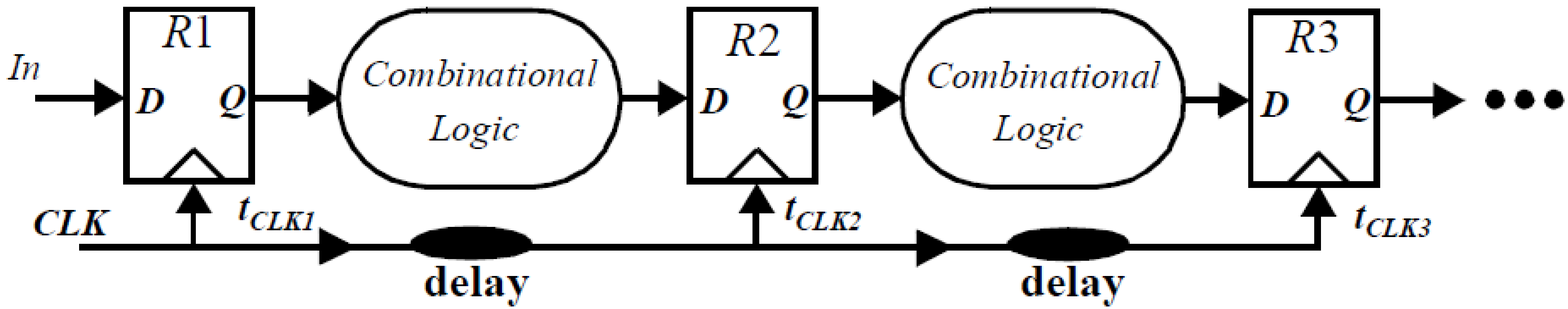


Clock Uncertainties

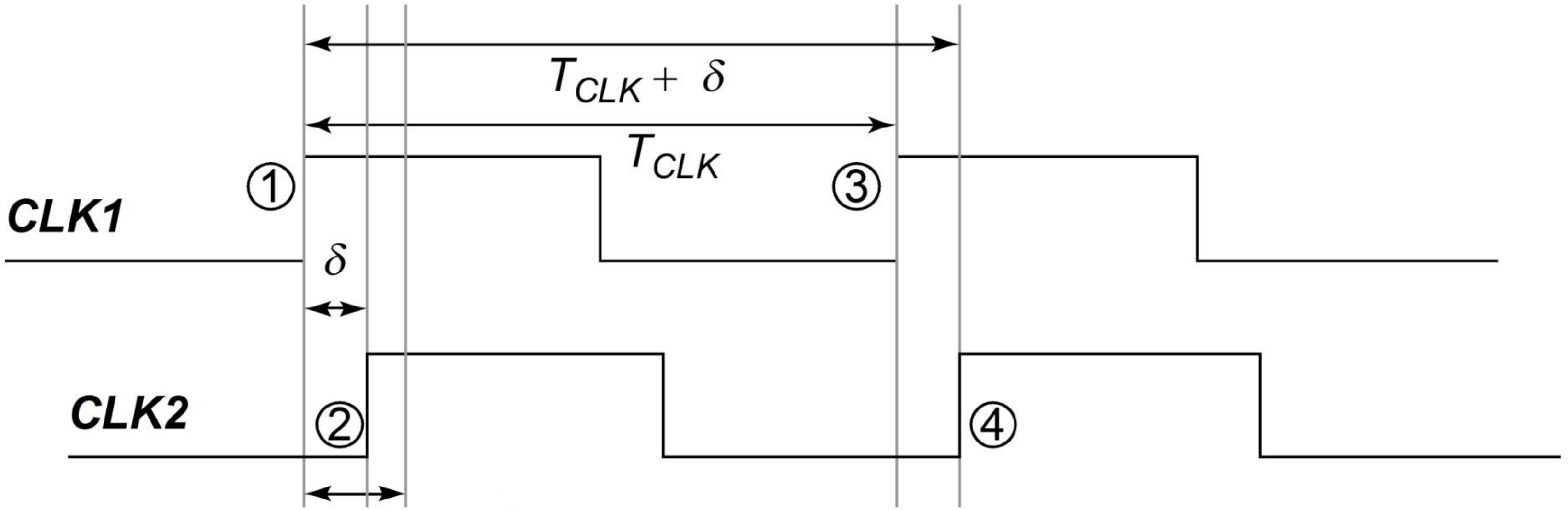
Interconnect Variations (3)



Positive and Negative Skew

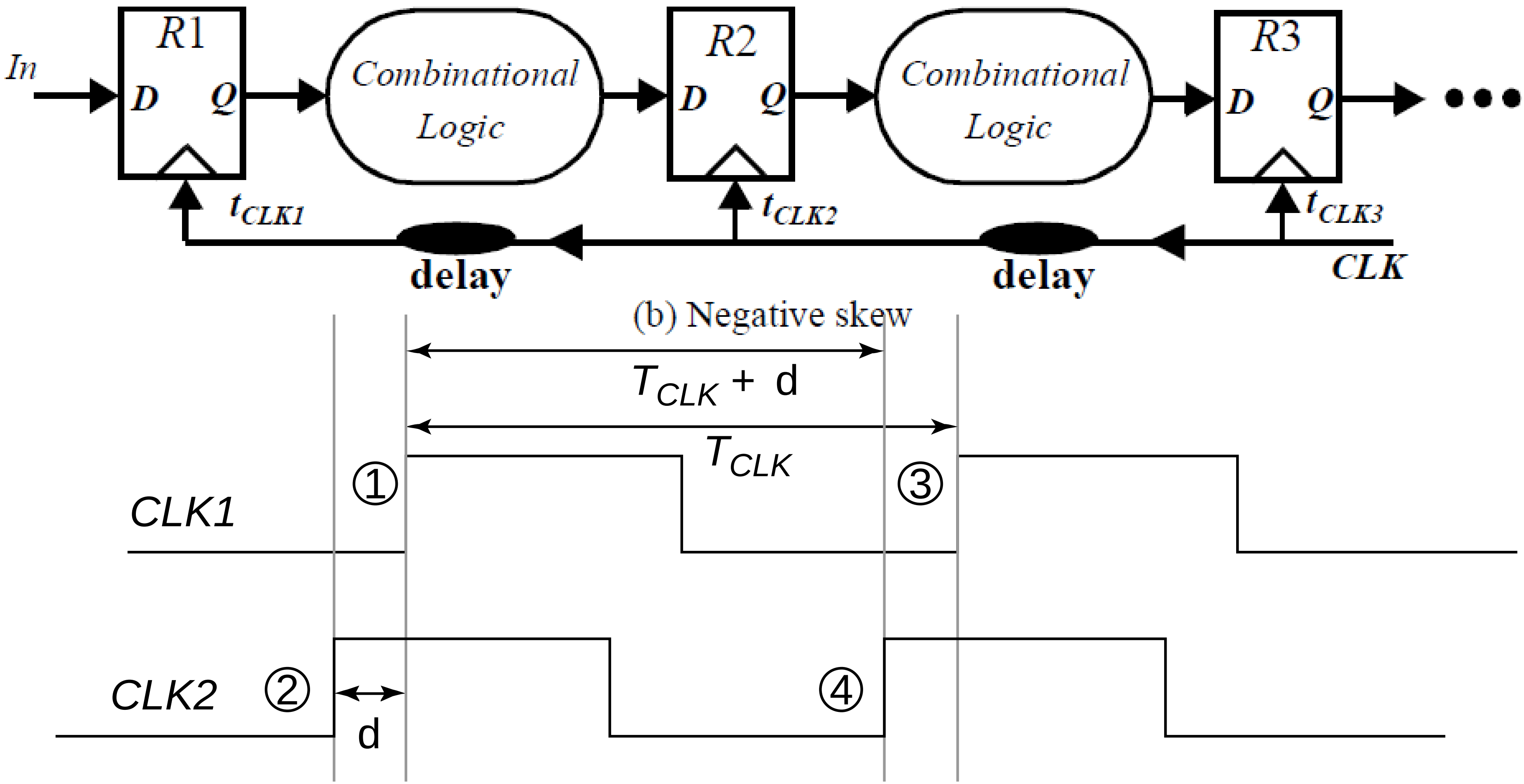


(a) Positive skew



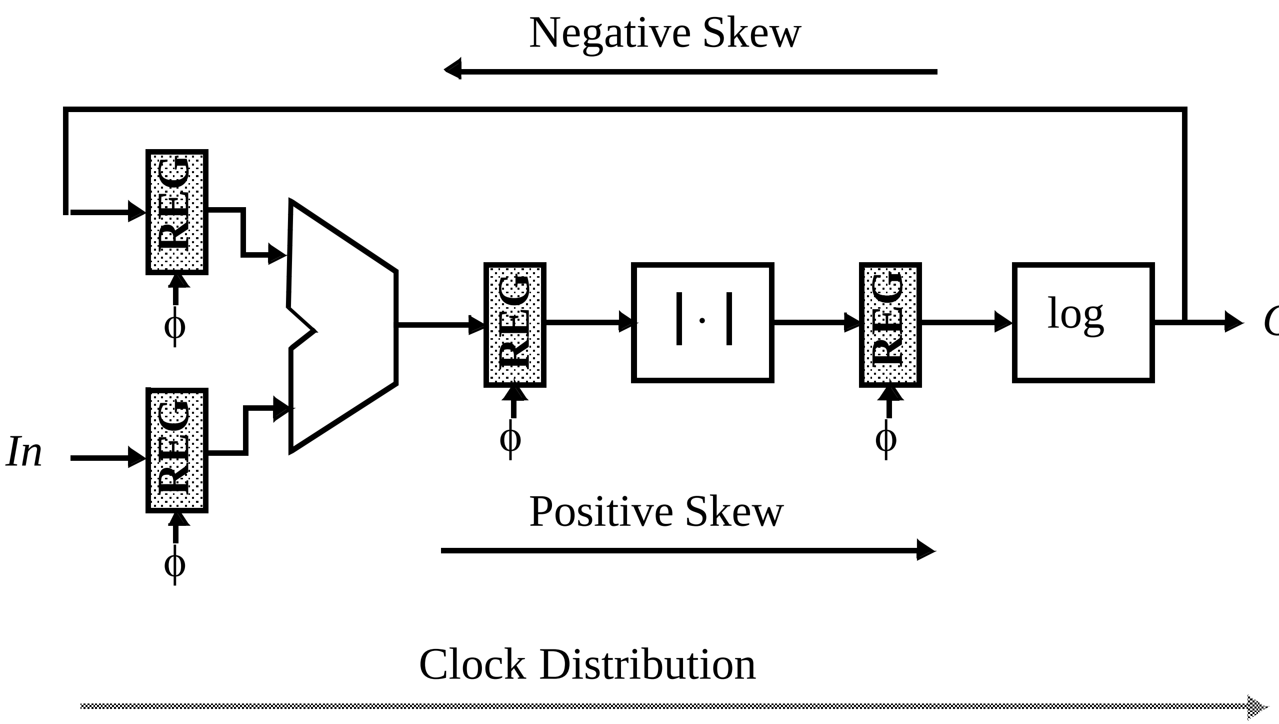
Launching edge arrives before the receiving edge

Negative Skew



Receiving edge arrives before the launching edge

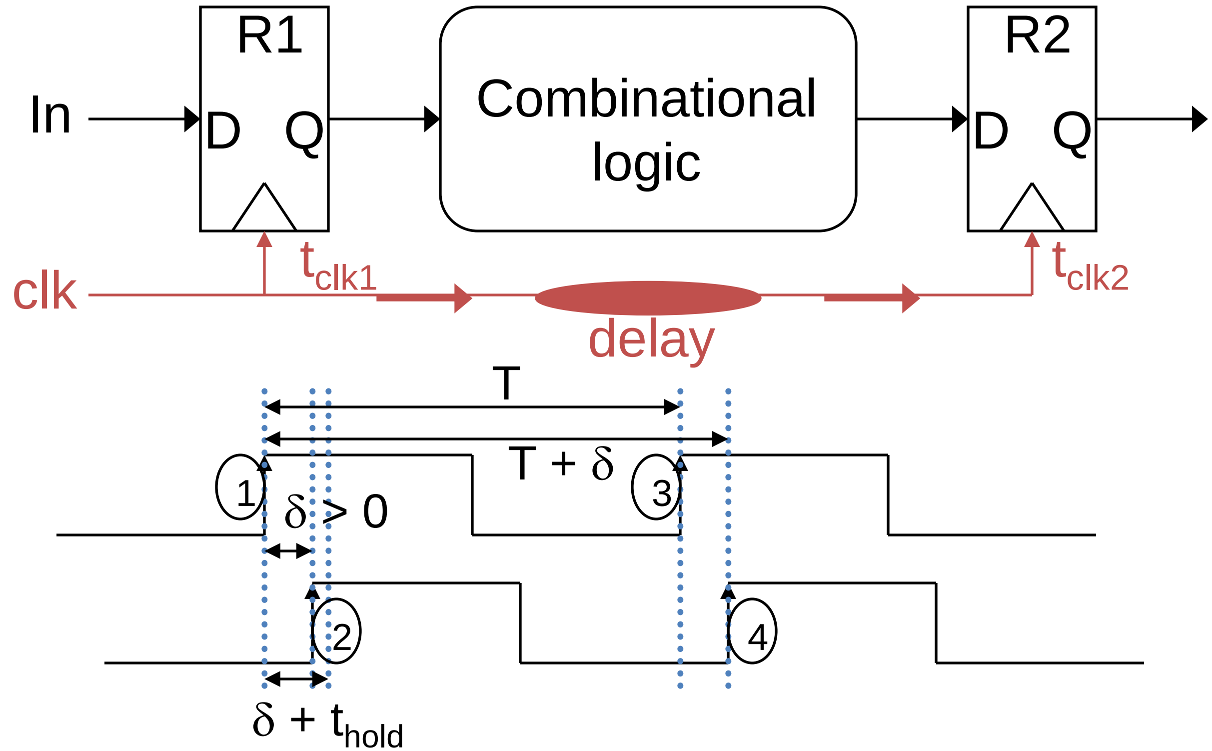
How to counter Clock Skew?



Data and Clock Routing

Positive Clock Skew

- Clock and data flow in the same direction



T : $T + \delta \geq t_{c\text{-}q} + t_{plogic} + t_{su}$ so $T \geq t_{c\text{-}q} + t_{plogic} + t_{su} - \delta$

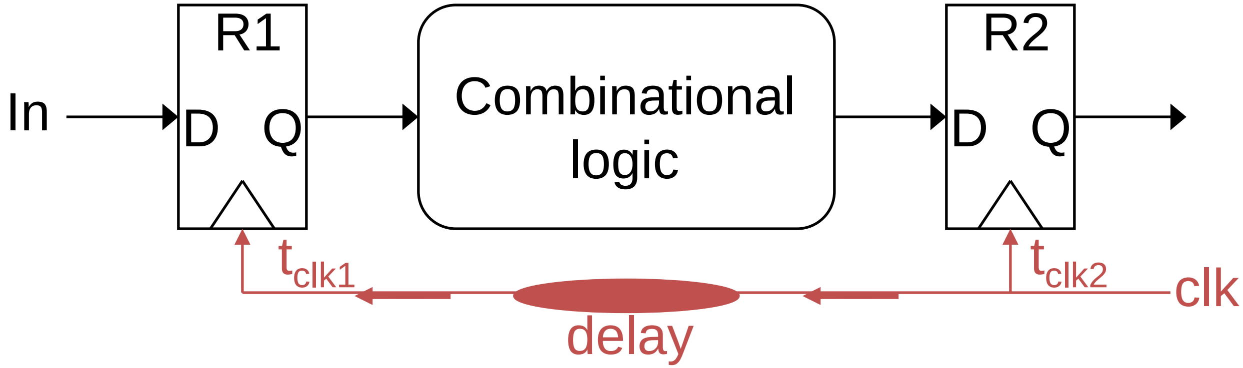
t_{hold} : $t_{hold} + \delta \leq t_{cdlogic} + t_{cdreg}$ so $t_{hold} \leq t_{cdlogic} + t_{cdreg} - \delta$

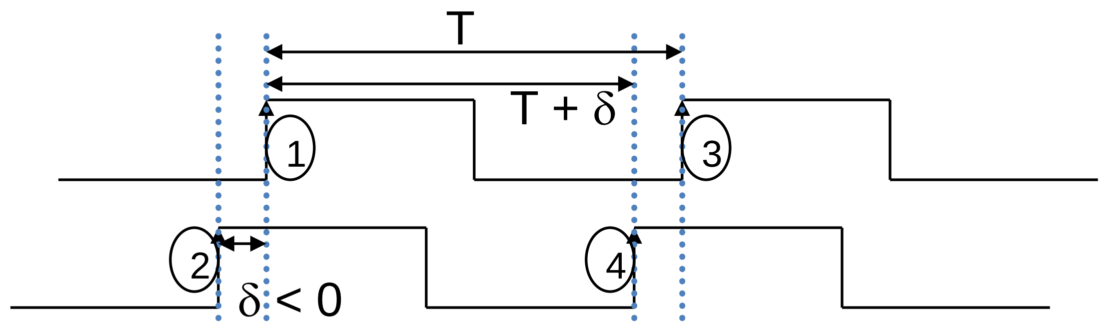
$\delta \leq tcdlogic + tcdreg - thold$

- $\delta > 0$: Improves performance, but makes t_{hold} harder to meet. If t_{hold} is not met (race conditions), the circuit malfunctions independent of the clock period!

Negative Clock Skew

- Clock and data flow in opposite directions





T : $T + \delta \geq t_{c\text{-}q} + t_{plogic} + t_{su}$ so $T \geq t_{c\text{-}q} + t_{plogic} + t_{su} - \delta$

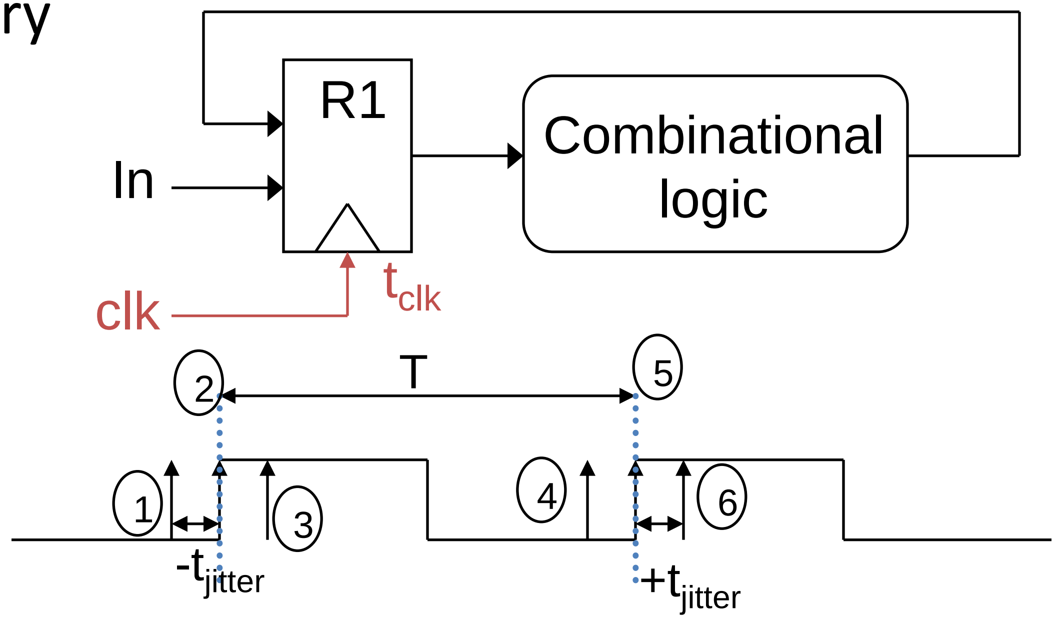
t_{hold} : $t_{hold} + \delta \leq t_{cdlogic} + t_{cdreg}$ so $t_{hold} \leq t_{cdlogic} + t_{cdreg} - \delta$

$\delta \leq t_{cdlogic} + t_{cdreg} - t_{hold}$

- $\delta < 0$: Degrades performance, but t_{hold} is easier to meet (eliminating race conditions)

Clock Jitter

- Jitter causes T to vary on a cycle-by-cycle basis

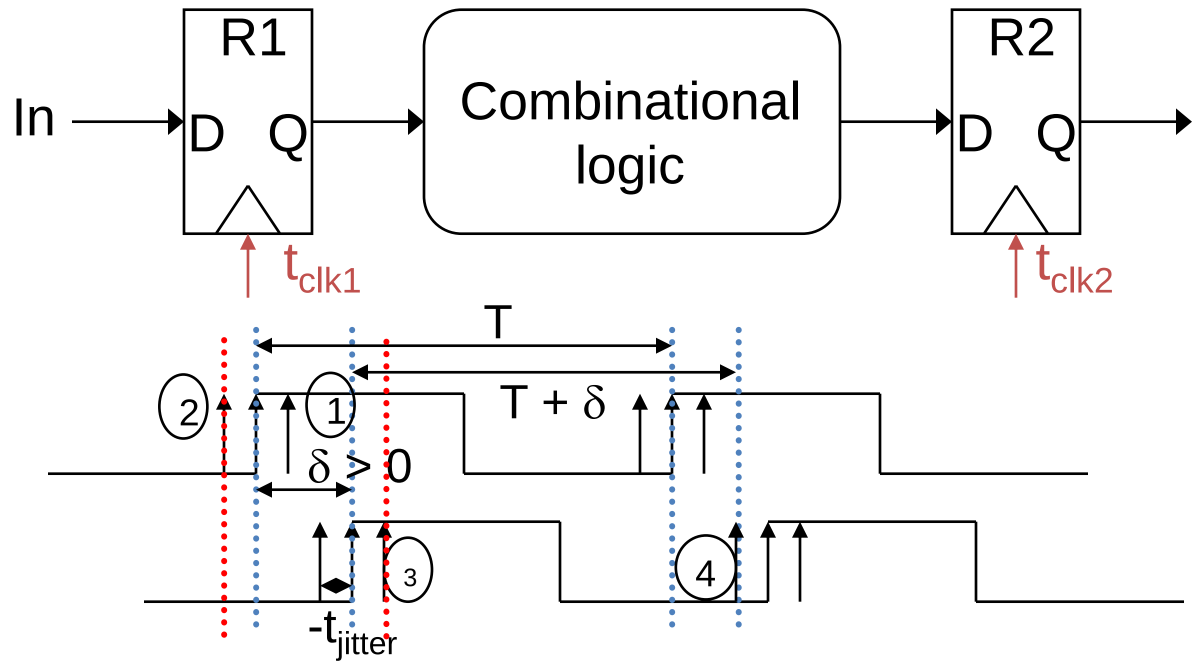


T : $T - 2t_{jitter} \geq t_{c-q} + t_{plogic} + t_{su}$ so $T \geq t_{c-q} + t_{plogic} + t_{su} + 2t_{jitter}$

- Jitter directly reduces the performance of a sequential circuit

Combined Impact of Skew and Jitter

- Constraints on the minimum clock period ($\delta > 0$)



$$T \geq t_{c\text{-}q} + t_{plogic} + t_{su} - \delta + 2t_{jitter} \qquad t_{hold} + \delta + 2t_{jitter} \leq t_{cdlogic} + t_{cdreg}$$

- $\delta > 0$ with jitter: Degrades performance, and makes t_{hold} even *harder* to meet. (The acceptable skew is reduced by jitter.)

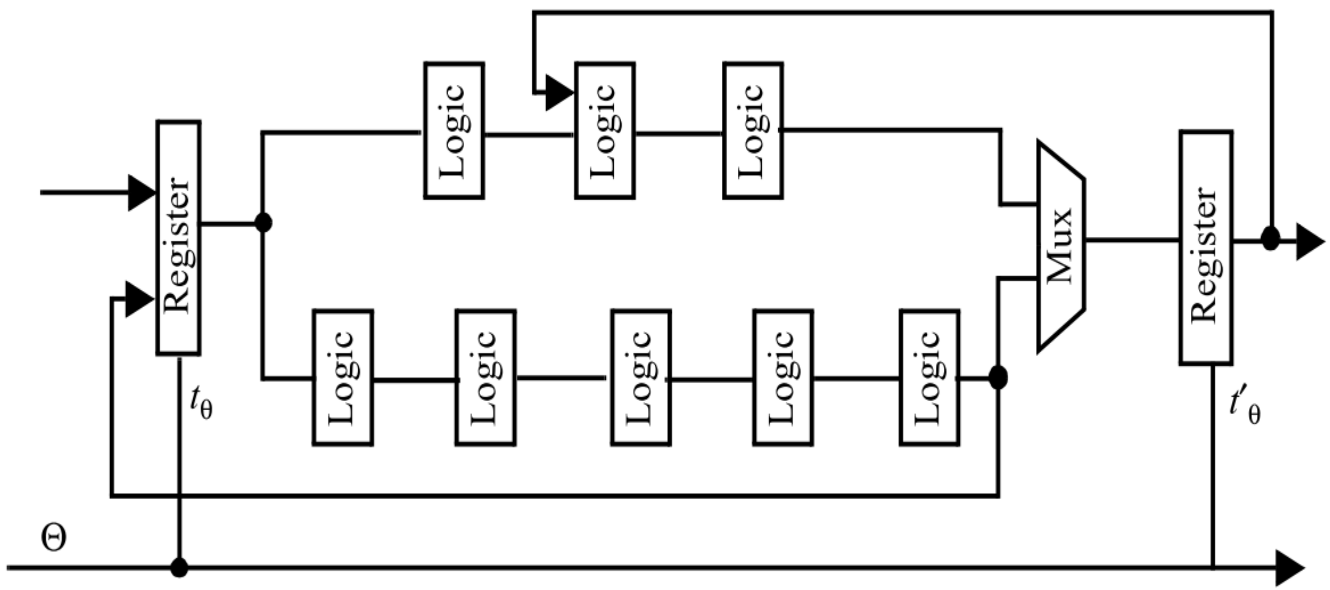
Example

9-1. For the circuit in figure below, assume a unit delay through the Register and Logic blocks (i.e., $t_R = t_L = 1$). Assume that the registers, which are positive edge-triggered, have a set-up time t_S of 1. The delay through the multiplexer t_M equals $2t_R$.

a. Determine the minimum clock period. Disregard clock skew.

b. Repeat part a, factoring in a non-zero clock skew: $\delta = t'_\theta - t_\theta = 1$.

c. Repeat part a, factoring in a non-zero clock skew: $\delta = t'_\theta - t_\theta = 4$.



a. Determine the minimum clock period. Disregard clock skew.

Solution

The circuit and paths of interest has been reproduced for convenience in Figure 0.1

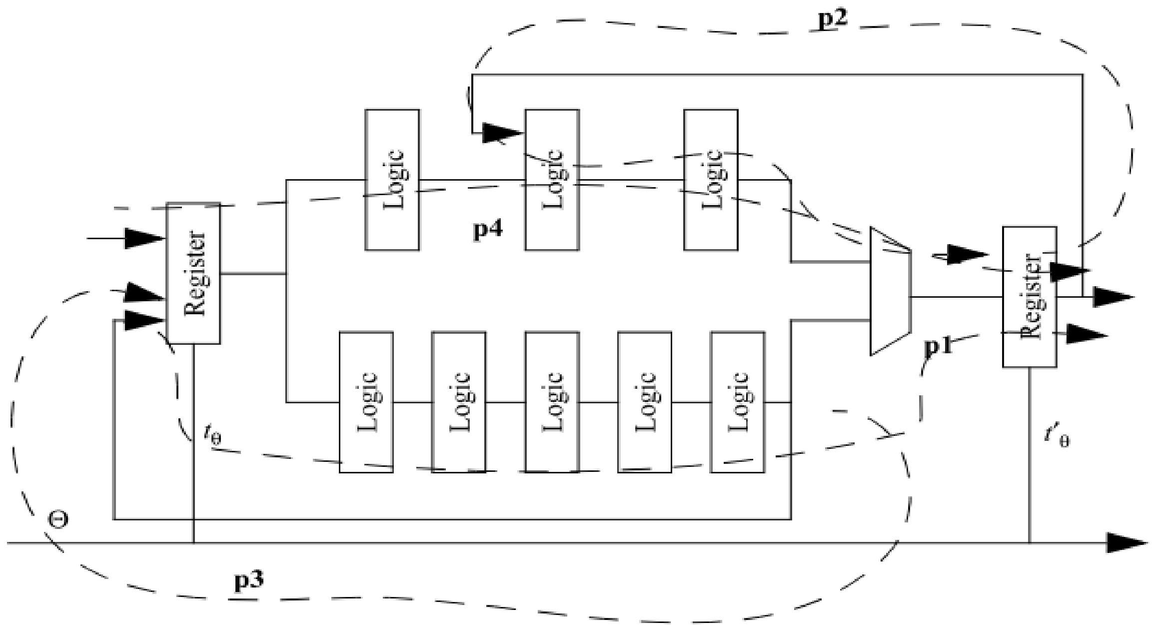


Figure 0.1 Sequential circuit.

Out of the 4 paths shown in the figure, p1 is the critical one and determines the lower bound on the clock period. Using $T \geq t_{reg} + t_{logic} + t_{setup} - \delta$, we get $\boldsymbol{T_{min}}$ **= 1+7+1 = 9**.

b. Repeat part a, factoring in a nonzero clock skew: $\delta = t'_\theta - t_\theta = 1$.

Solution

With finite clock skew, the time periods for different paths are as follows : $T_{min}(p1) = 9 - 1 = 8$, $T_{min}(p2) = 6$, $T_{min}(p3) = 7$, $T_{min}(p4) = 7 - 1 = 6$ (Note that the clock skew is 0 for paths p2 and p3). Therefore the minimum clock period is $\boldsymbol{T_{min}}$ **= 8**.

c. Repeat part a, factoring in a non-zero clock skew: $\delta = t'_\theta - t_\theta = 4$.

Solution

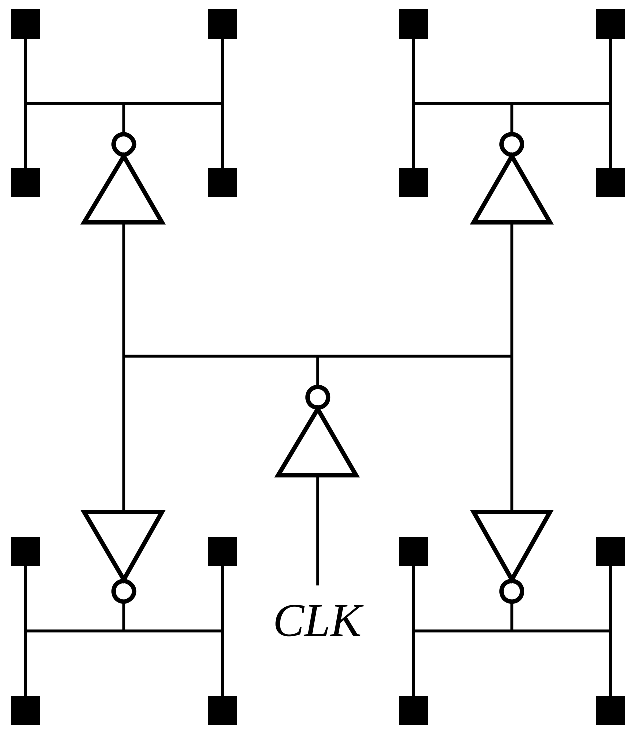
As the clock skew increases, the most significant path changes. Repeating the calculations in part (b) we get : $T_{min}(p1) = 9 - 4 = 5$, $T_{min}(p2) = 6$, $T_{min}(p3) = 7$, $T_{min}(p4) = 7 - 4 = 3$(Note that the clock skew is 0 for paths p2 and p3). Therefore the minimum clock period is $\boldsymbol{T_{min}}$ **= 7**.

Clock Distribution Networks

- Clock skew and jitter can ultimately limit the performance of a digital system, so designing a clock network that minimizes both is important
  - In many high-speed processors, a majority of the dynamic power is dissipated in the clock network
  - To reduce dynamic power, the clock network must support clock gating (shutting down (disabling the clock) units)
- Clock distribution techniques
  - Balanced paths (H-tree network, matched RC trees)
    - In the ideal case, can eliminate skew
    - Could take multiple cycles for the clock signal to propagate to the leaves of the tree
  - Clock grids
    - Typically used in the final stage of the clock distribution network
    - Minimizes absolute delay (not relative delay)

Clock Distribution

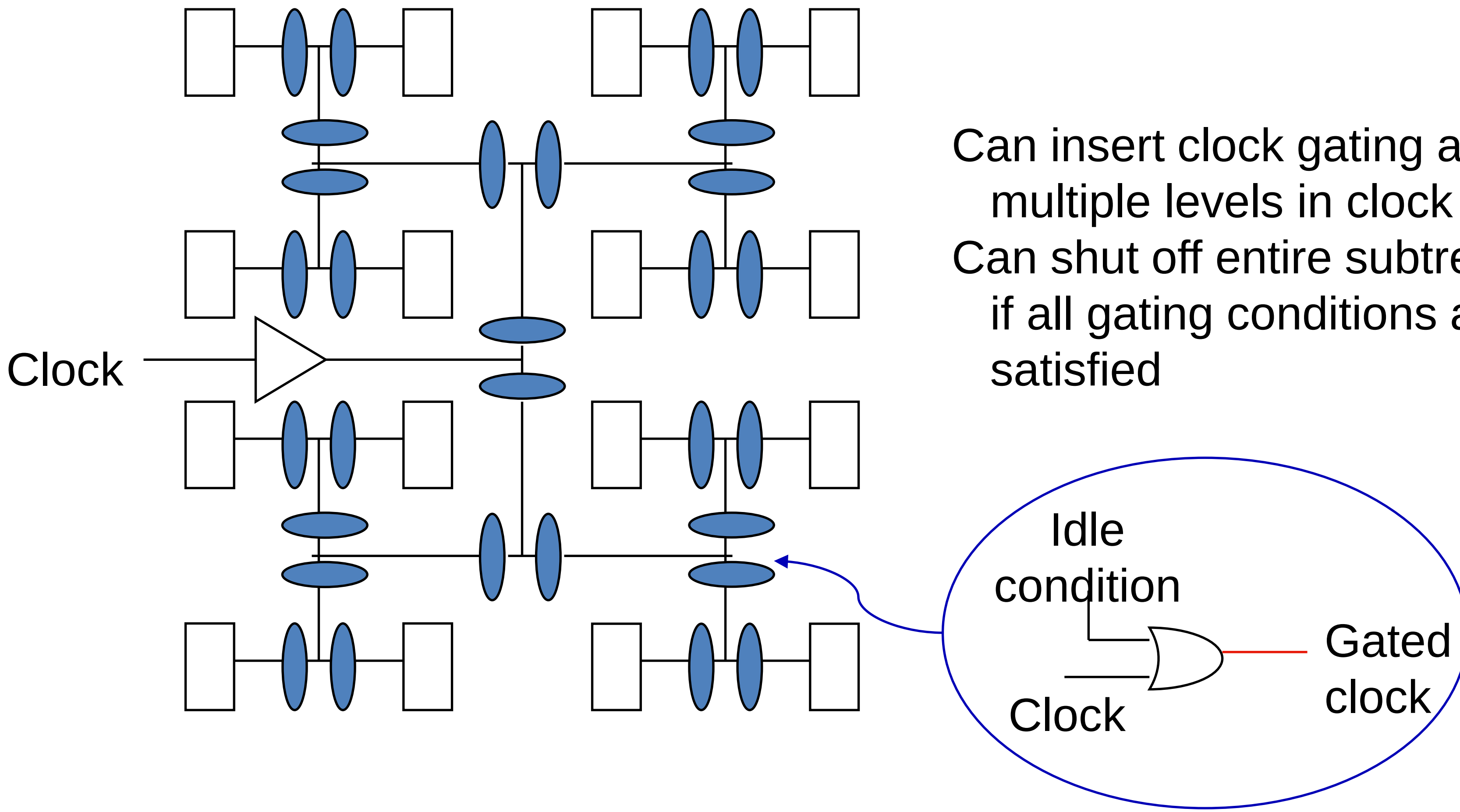
H-tree



Clock is distributed in a tree-like fashion

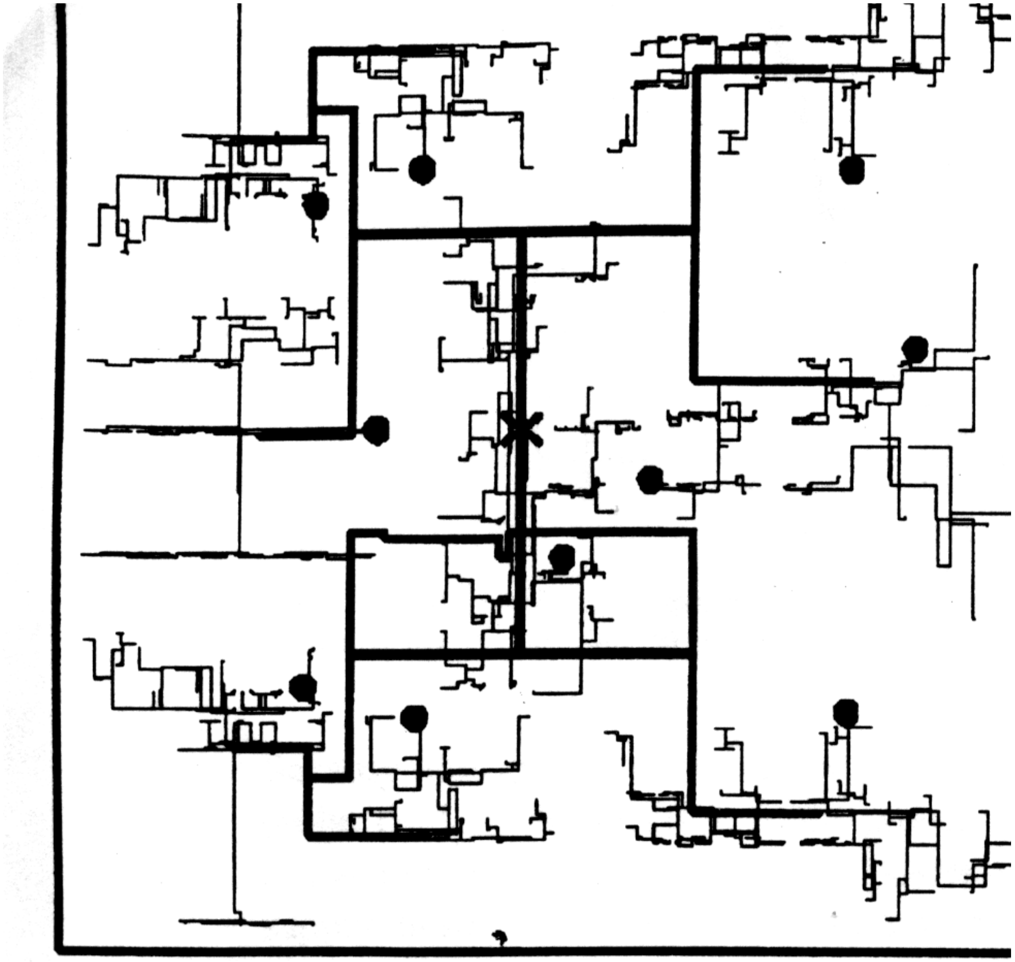
H-Tree Clock Network

- If the paths are perfectly balanced, clock skew is zero



Can insert clock gating at multiple levels in clock tree
Can shut off entire subtree if all gating conditions are satisfied

More realistic H-tree



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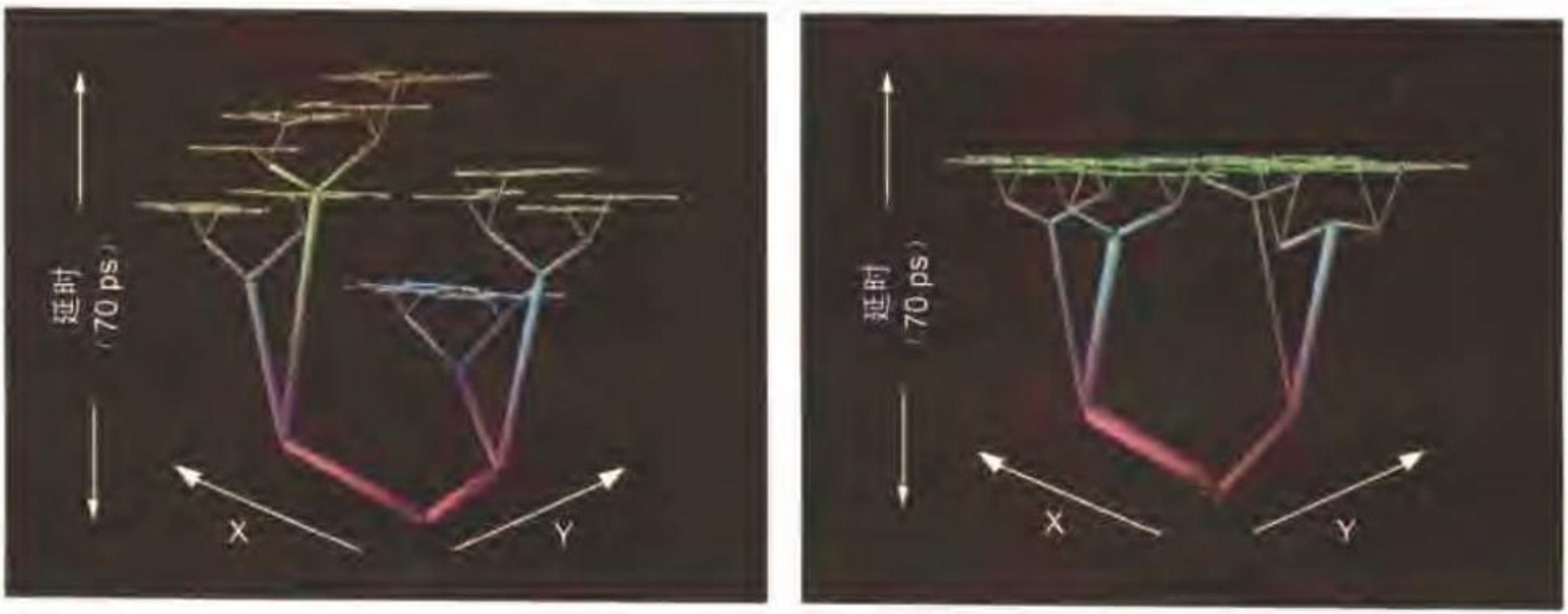
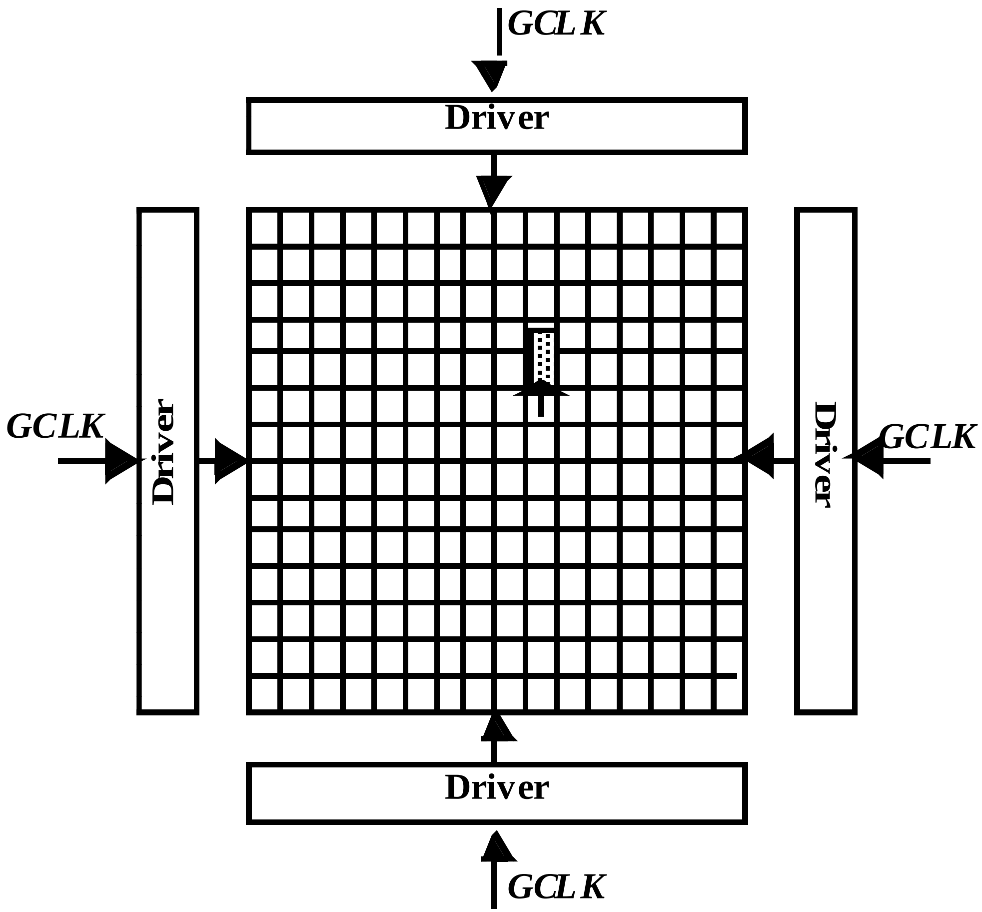

延时
（70 ps）
X
Y
延时
（70 ps）
X
Y

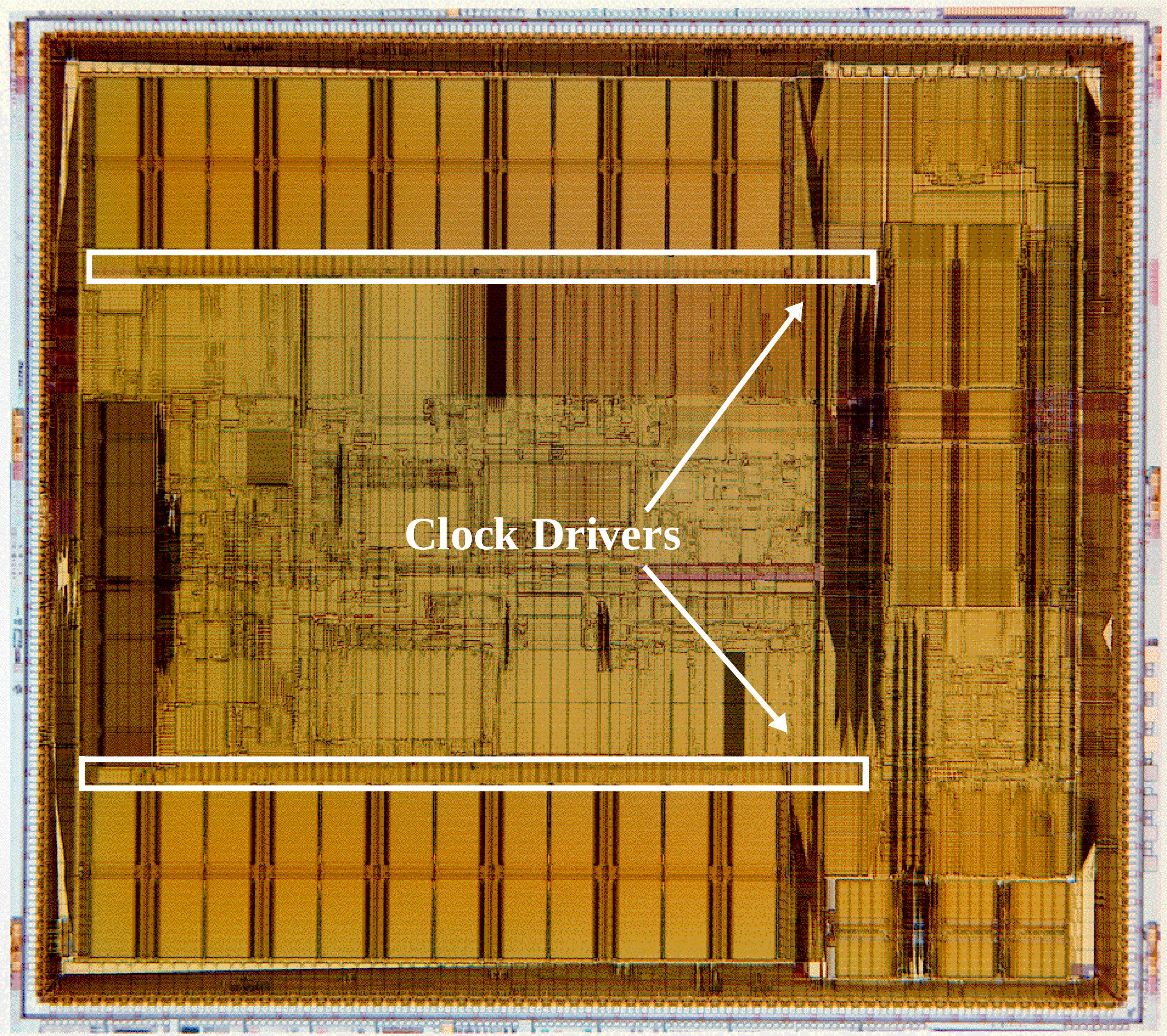
The Grid System



- No rc-matching
- Large power

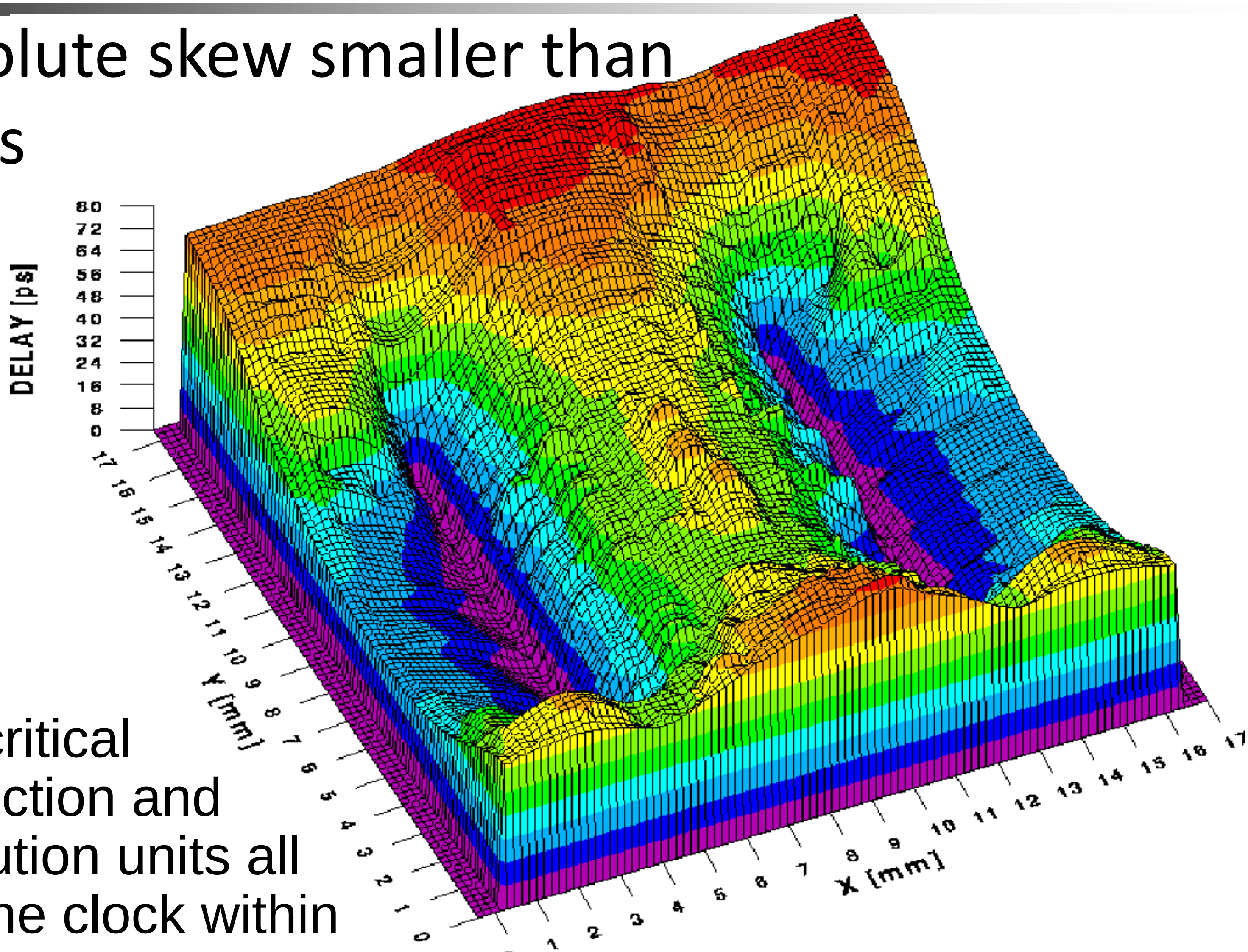
DEC Alpha 21164 (EV5)

- 300 MHz clock (9.3 million transistors on a 16.5x18.1 mm die in 0.5 micron CMOS technology)
  - single phase clock
- 3.75 nF total clock load
  - Extensive use of dynamic logic
- 20 W (out of 40) in clock distribution network
- Two level clock distribution
  - Single 6 stage driver at the center of the chip
  - Secondary buffers drive the left and right sides of the clock grid in m3 and m4
- Total equivalent driver size of 58 cm !!


Clock Drivers

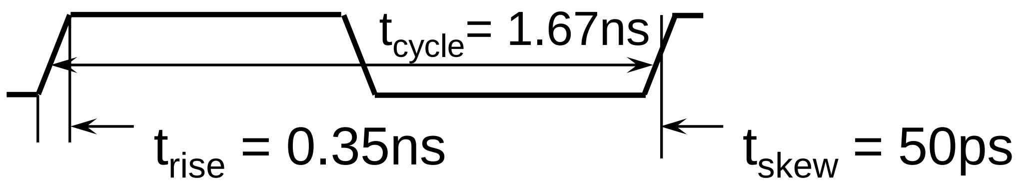
Clock Skew in Alpha Processor

- Absolute skew smaller than 90 ps

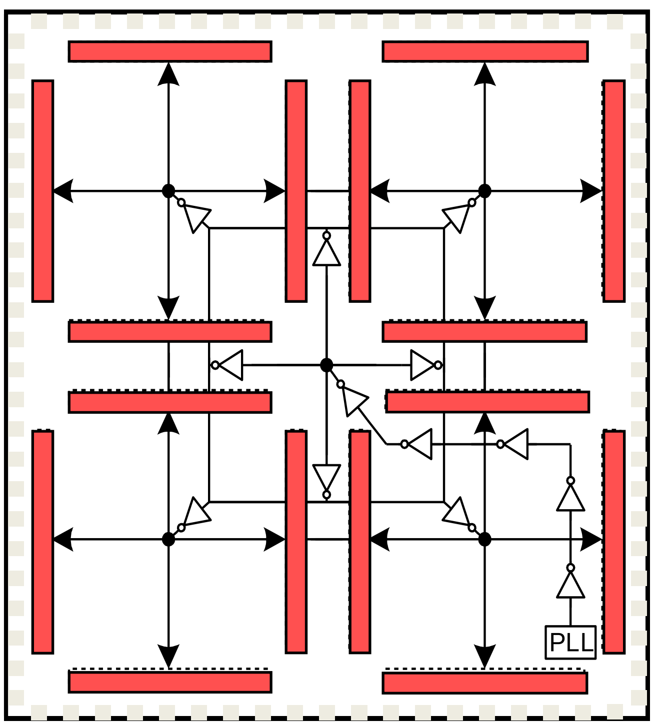


- The critical instruction and execution units all see the clock within 65 ps

EV6 (Alpha 21264) Clocking
600 MHz – 0.35 micron CMOS

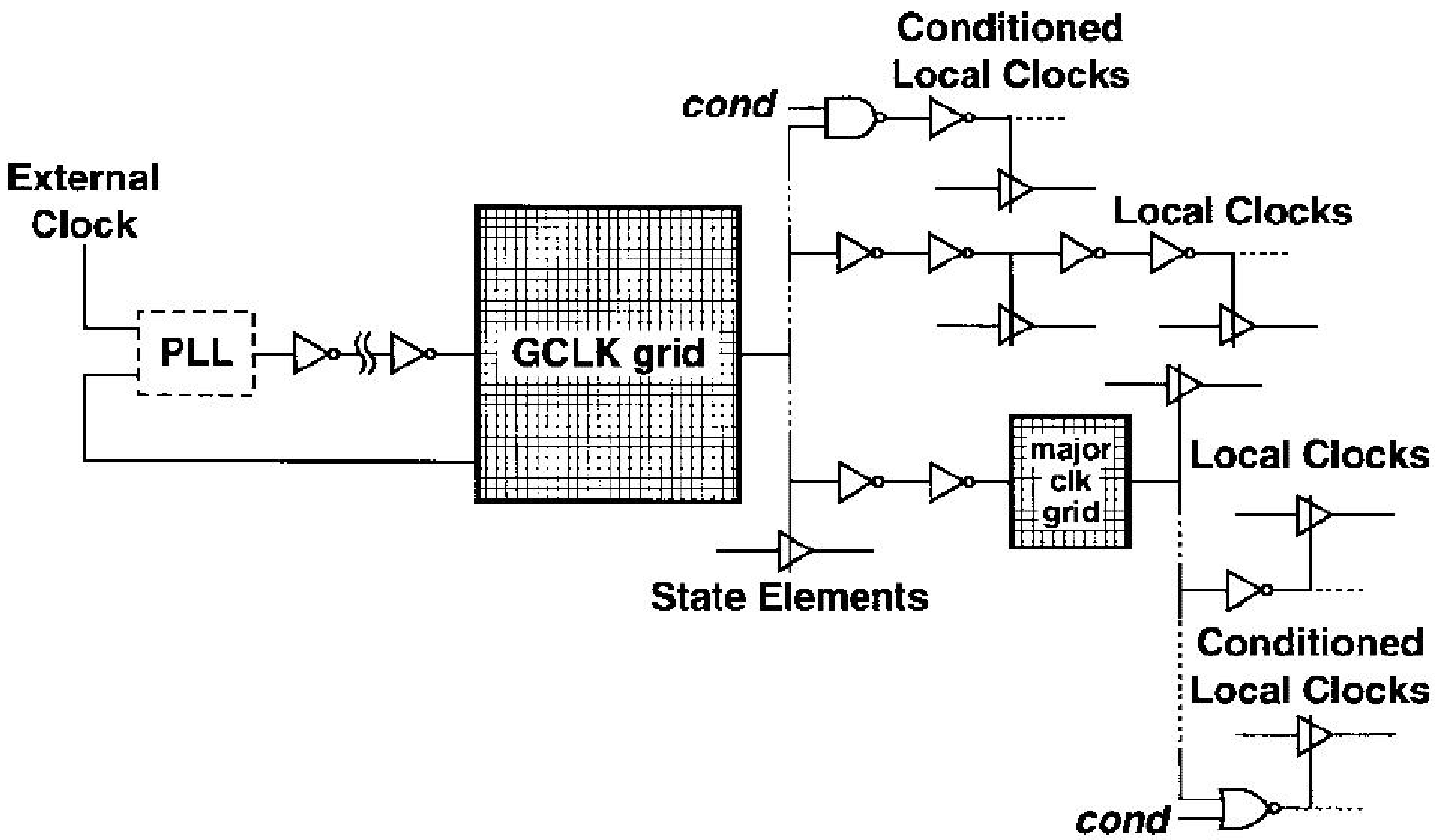


Global clock waveform

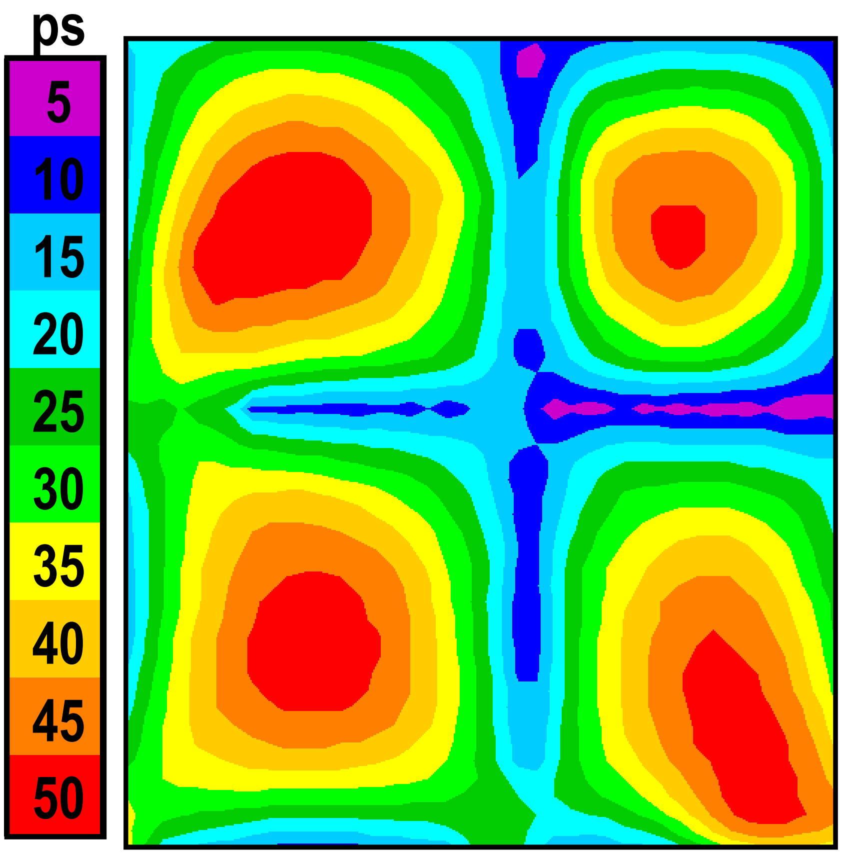


- 2 Phase, with multiple conditional buffered clocks
  - 2.8 nF clock load
  - 40 cm final driver width
- Local clocks can be gated “off” to save power
- Reduced load/skew
- Reduced thermal issues
- Multiple clocks complicate race checking

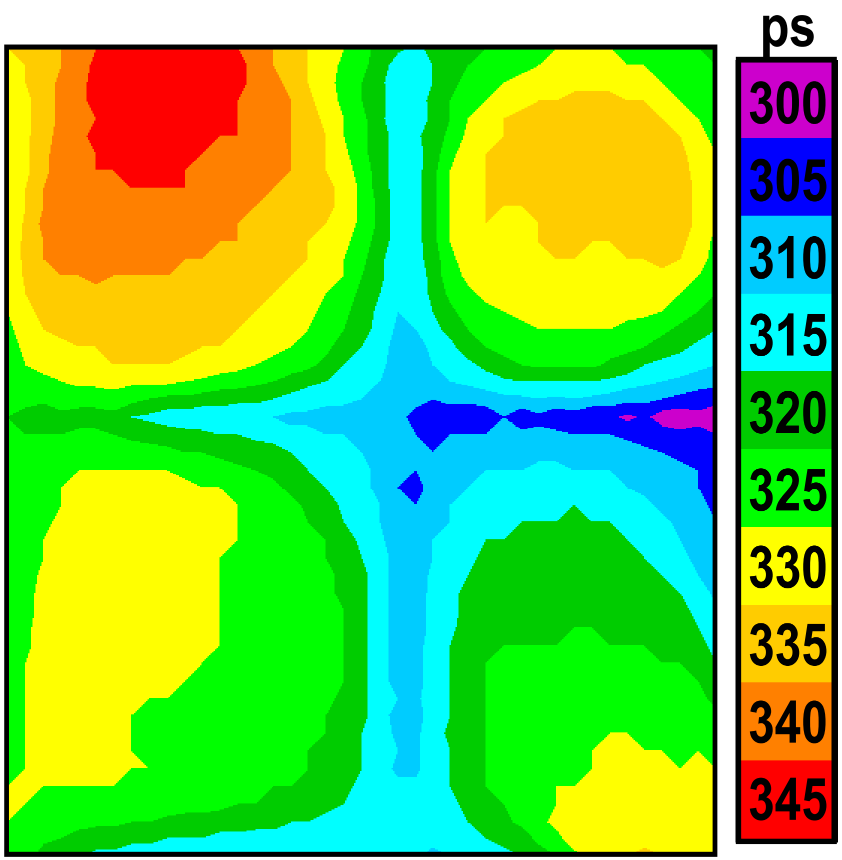
21264 Clocking



EV6 Clock Results



GCLK Skew
(at Vdd/2 Crossings)

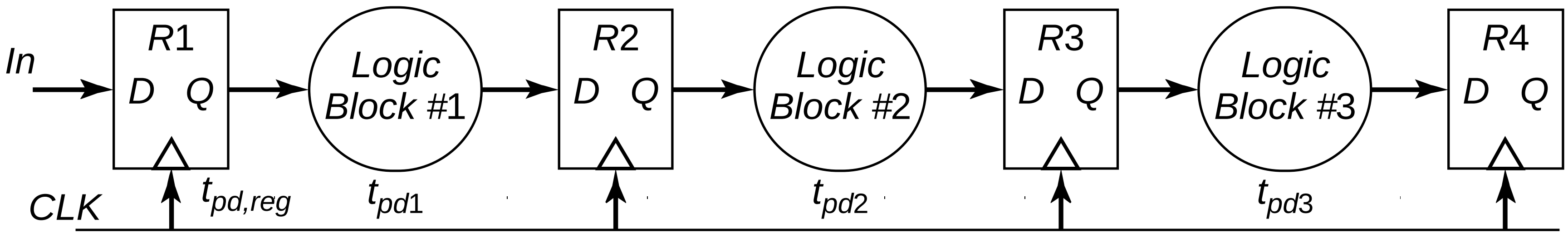


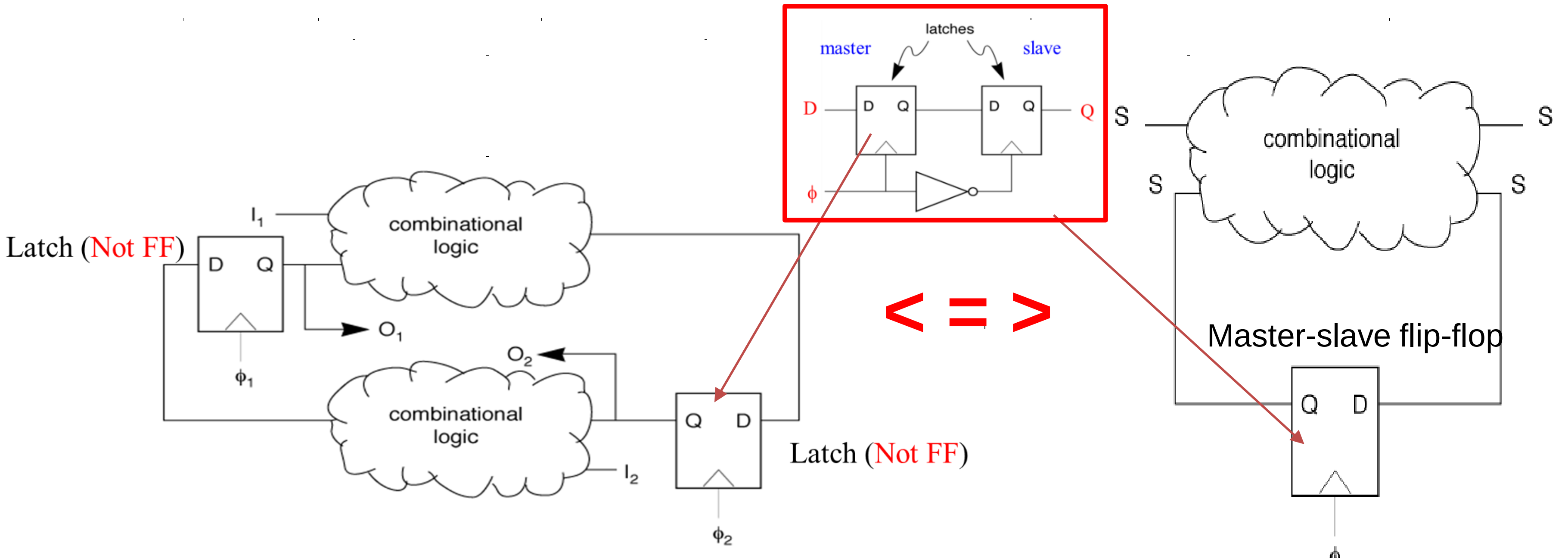
GCLK Rise Times
(20% to 80% Extrapolated to 0% to 100%)

Dealing with Clock Skew and Jitter

- To minimize skew, balance clock paths using *H-tree* or *matched-tree* clock distribution structures.
- If possible, route data and clock in opposite directions; eliminates races at the cost of performance.
- The use of gated clocks to help with dynamic power consumption make jitter worse.
- Shield clock wires (route power lines – V_{DD} or GND – next to clock lines) to minimize/eliminate coupling with neighboring signal nets.
- Use dummy fills to reduce skew by reducing variations in interconnect capacitances due to interlayer dielectric thickness variations.
- Beware of temperature and supply rail variations and their effects on skew and jitter.
- Power supply noise fundamentally limits the performance of clock networks.

Synchronous Pipelined Datapath





Latch based Strict two-phase sequential machines

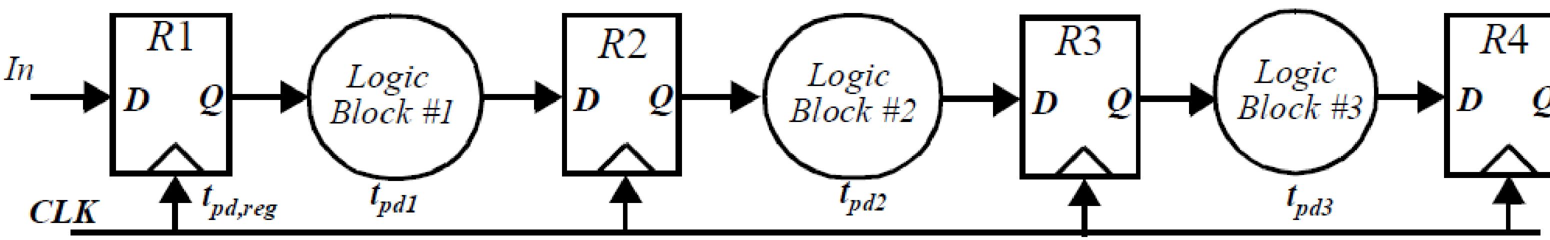
Flip-flop-based sequential machines

Self-timed and Asynchronous Design

Functions of clock in synchronous design

1) Acts as completion signal

2) Ensures the correct ordering of events



$$T > \max(t_{pd1}, t_{pd2}, t_{pd3}) + t_{pd,reg}$$

Asynchronous design

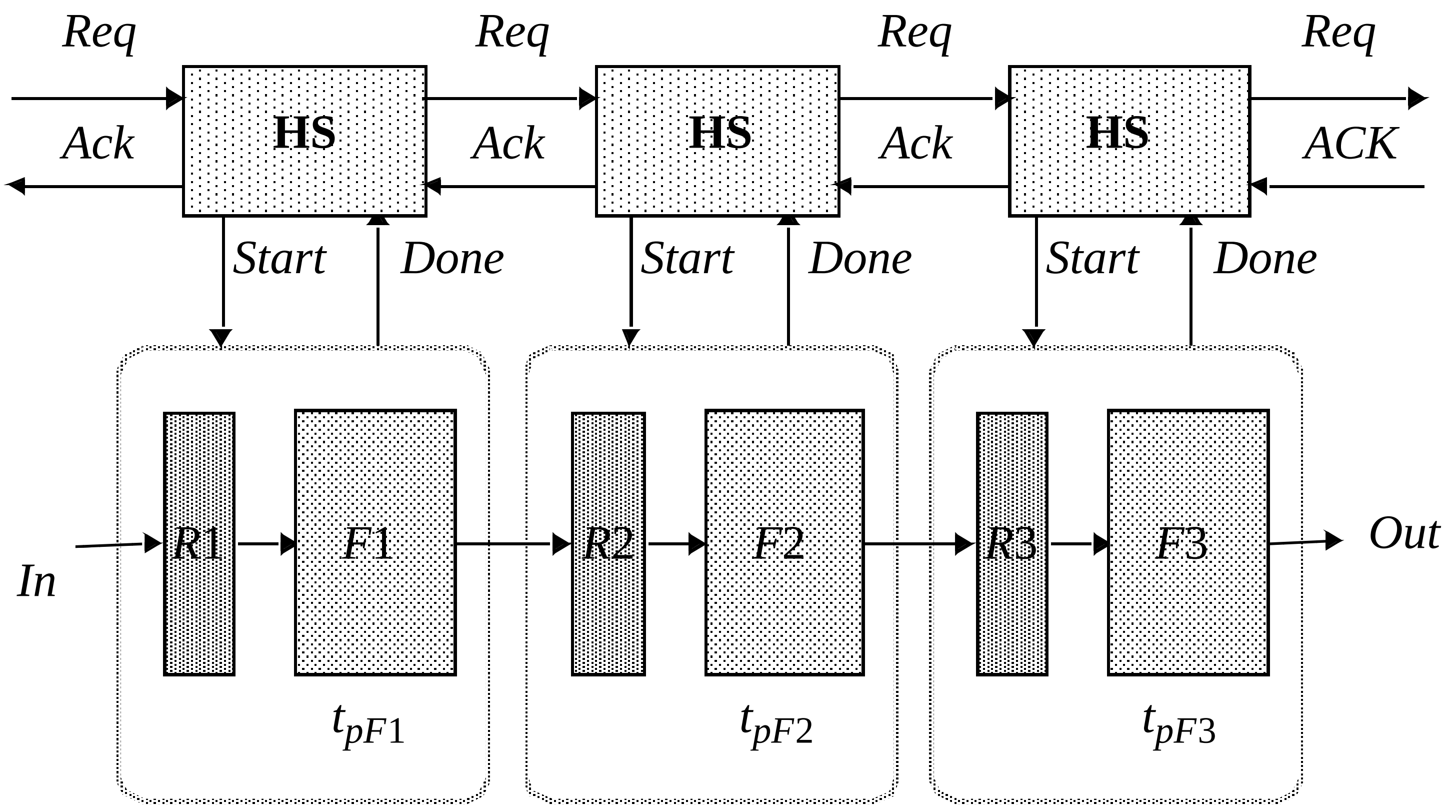
1) Completion is ensured by careful timing analysis

2) Ordering of events is implicit in logic

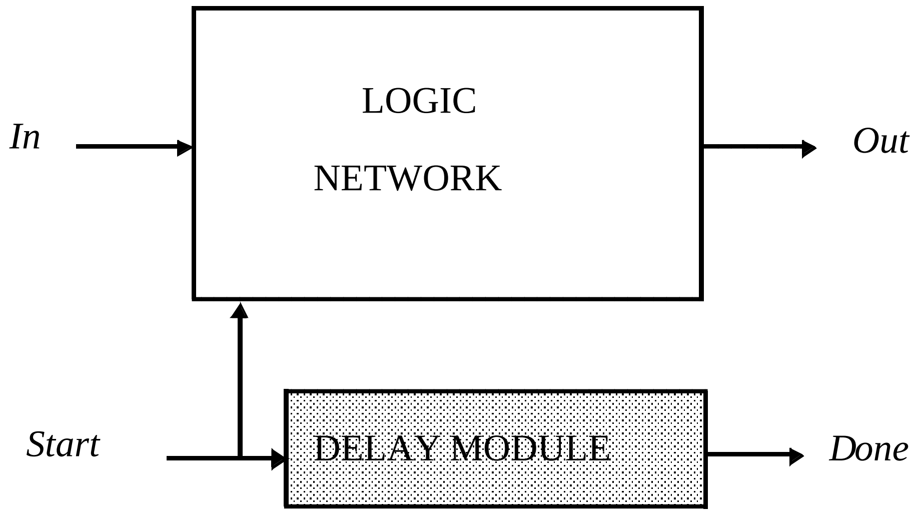
Self-timed design

Ordering imposed by handshaking protocol

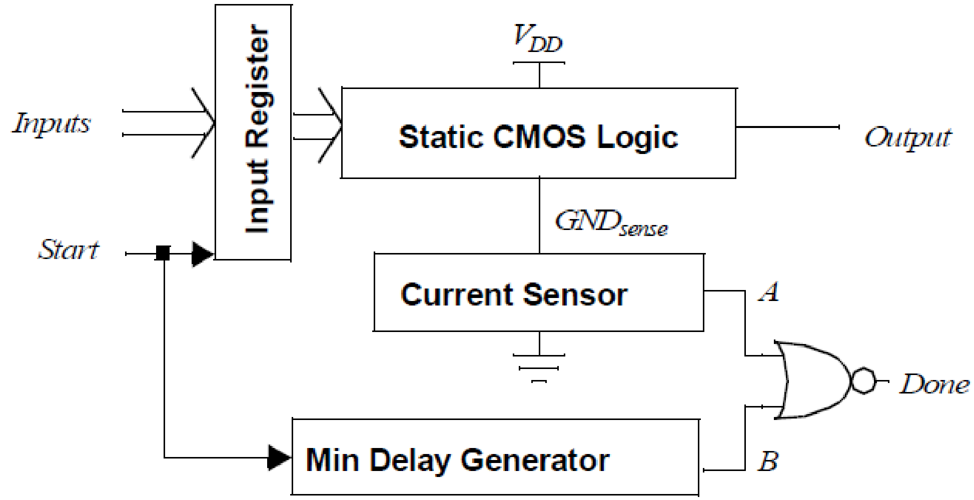
Self-Timed Pipelined Datapath



Completion Signal Generation

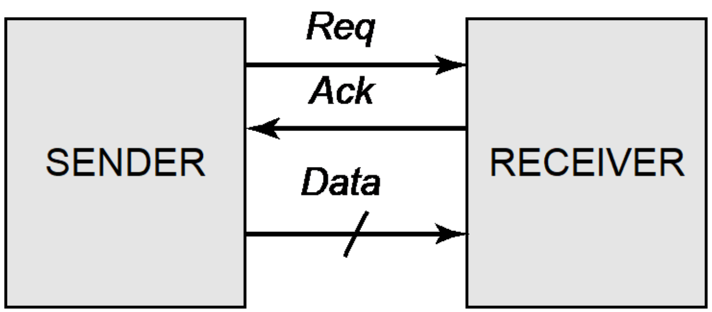


Using Delay Element (e.g. in memories)

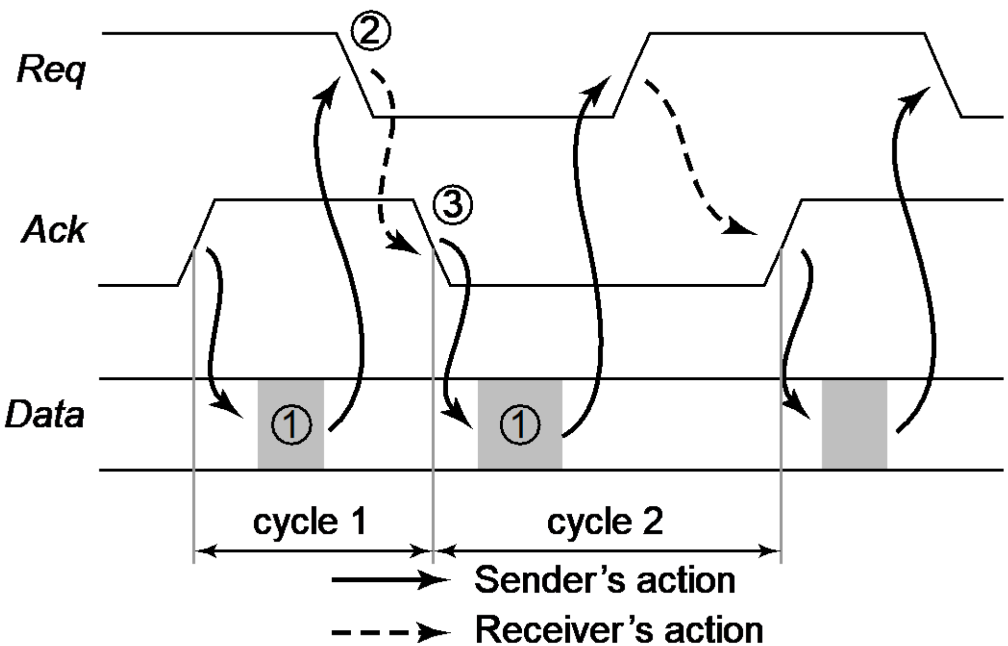


Completion-signal generation using current sensing

Hand-Shaking Protocol



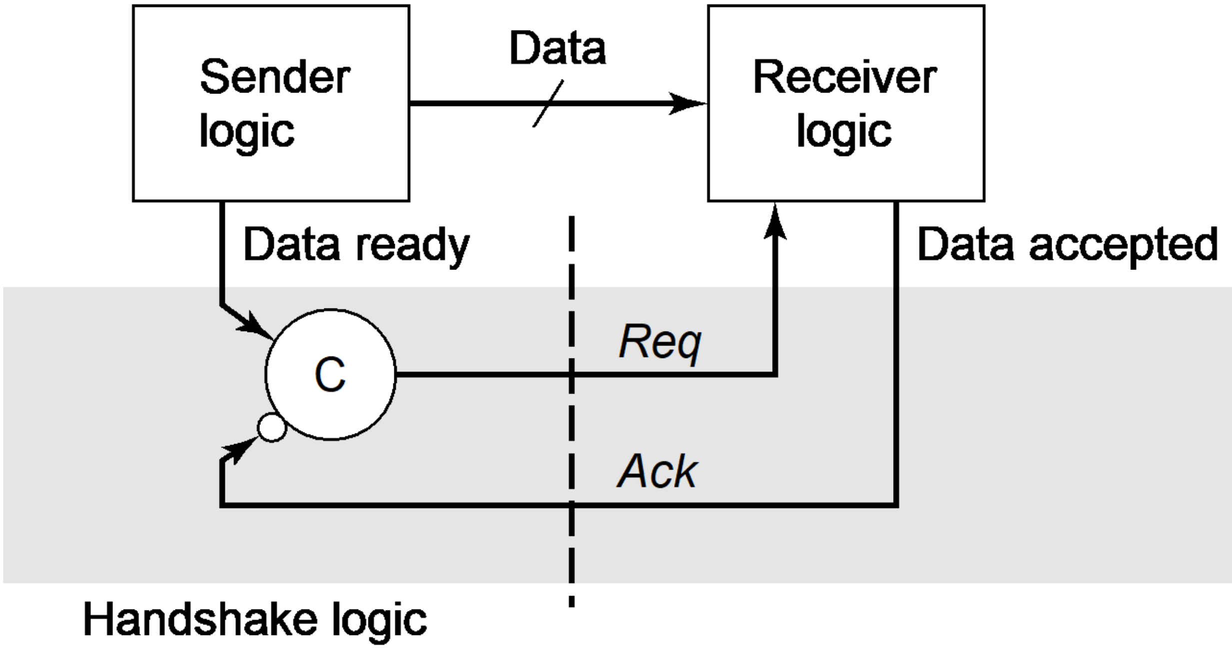
(a) Sender-receiver configuration



(b) Timing diagram

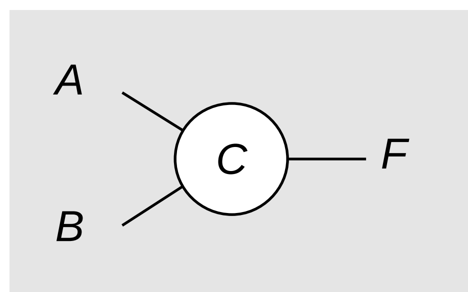
Two Phase Handshake

2-Phase Handshake Protocol



Advantage : FAST - minimal # of signaling events (important for global interconnect)

Disadvantage : edge - sensitive

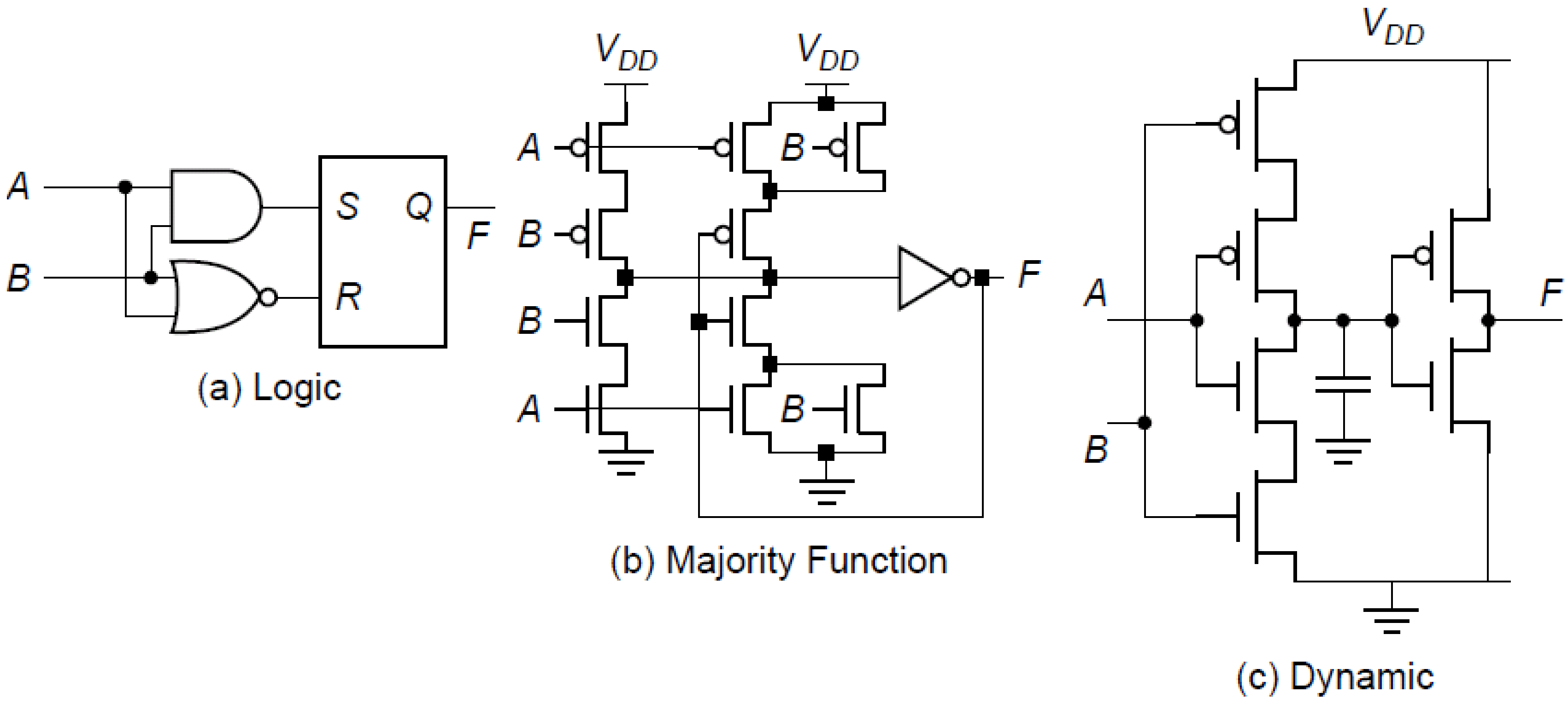


(a) Schematic

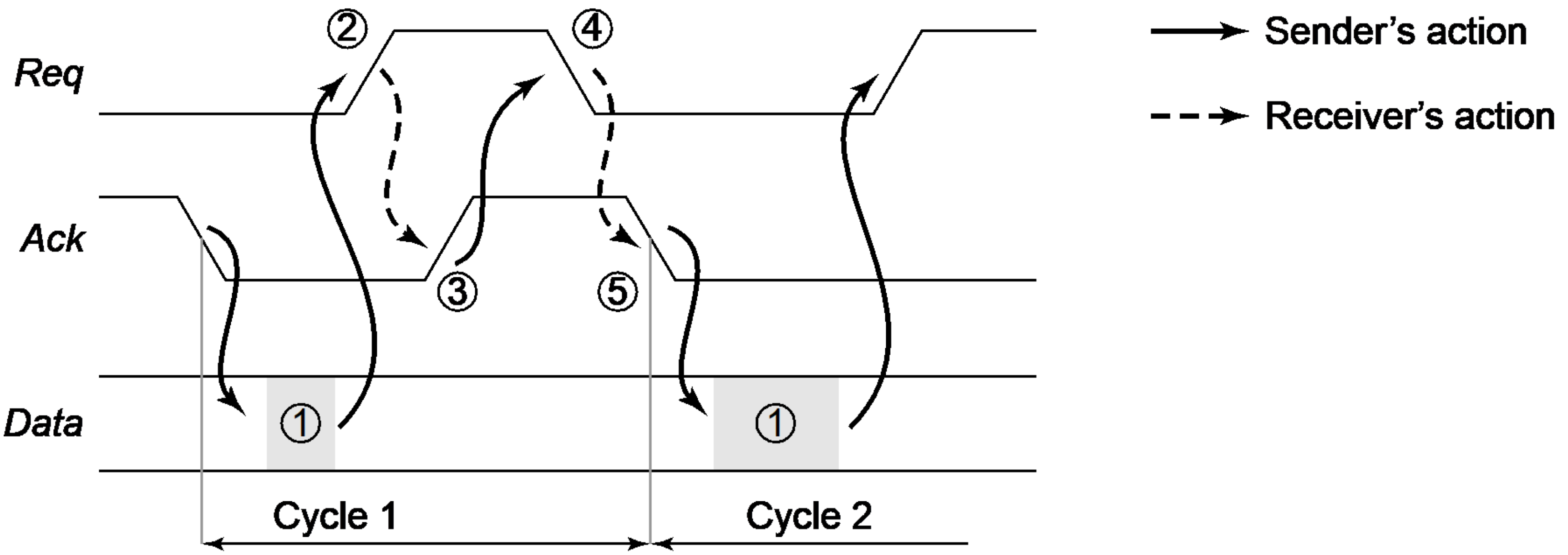
A	B	F_{n+1}
0	0	0
0	1	F_n
1	0	F_n
1	1	1

(b) Truth table

Muller C Circuit implementation



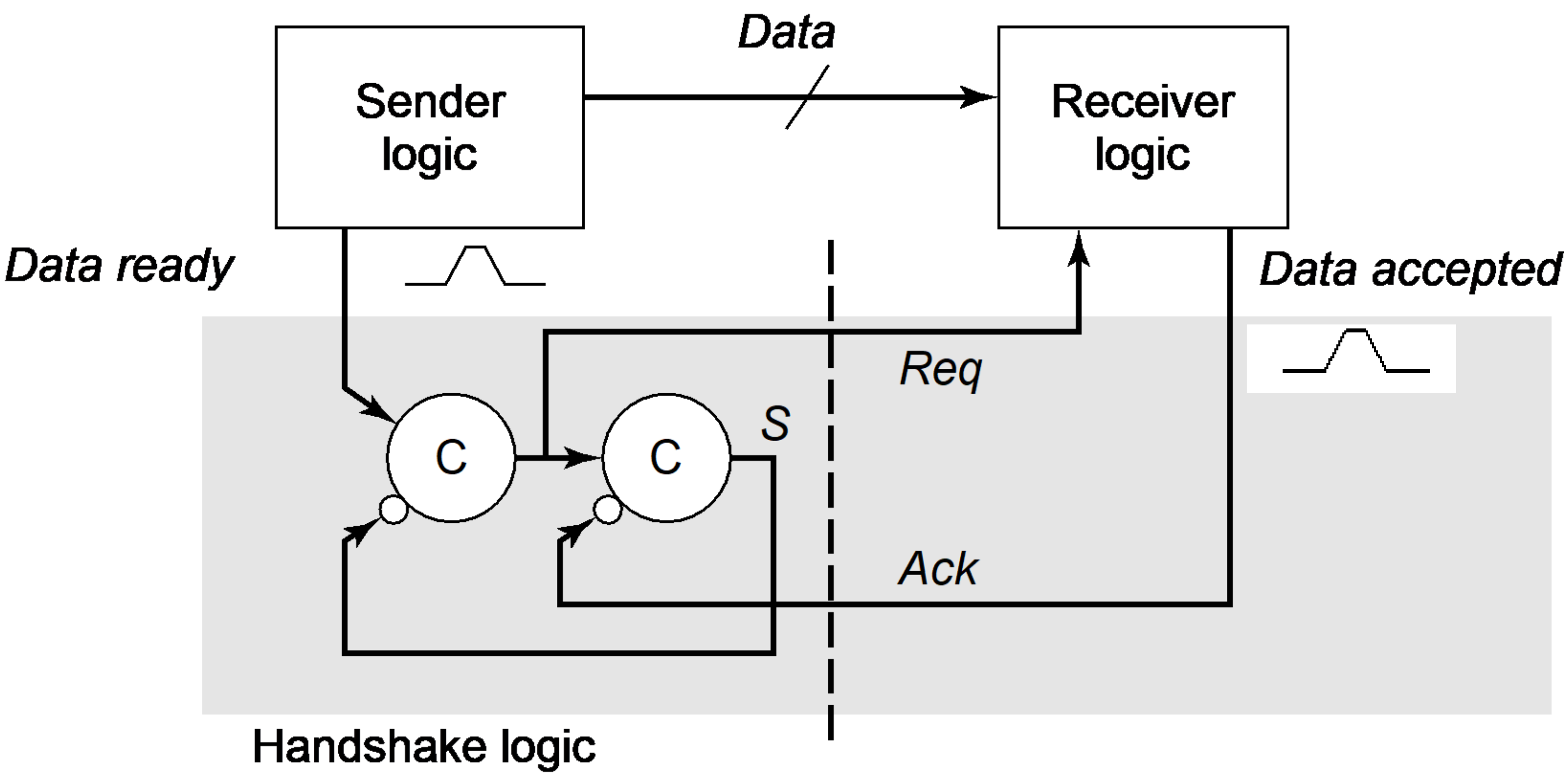
4-Phase Handshake Protocol



Slower, but unambiguous

4-Phase Handshake Protocol

Implementation using Muller-C elements

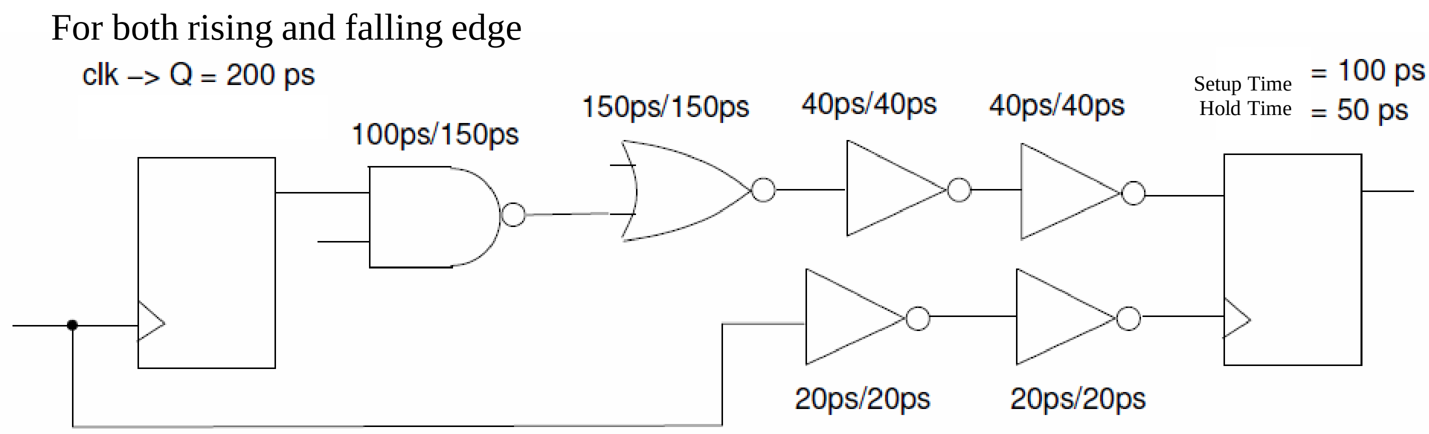


Summary

- Synchronous digital circuits
- Clock skew and jitter
- Clock distribution (H-tree, Grid)
- Asynchronous and self-timed design(Self-timed design handshaking protocol)

Homework 3

According to combinational logic in the figure below and through the static calculation, what is the maximum working frequency of the circuit? Pls also check whether the hold time satisfies timing constraints or not. [Remark: Annotations of the delays above the gates are "Rising Delay/Falling Delay", the memory elements are D flip-flops]



- Pls Send the electronic version of your homework before 23:59pm, Jan. 5th, 2025. THX!